



A comprehensive review of carbon dioxide capture, transportation, utilization, and storage: a source of future energy

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Abstract

Carbon dioxide (CO₂) is a harmful gas generated every day by burning fossil fuels in industry, automobiles, firing smoke, and so on. Excessive carbon dioxide creates global warming and severe health issues for humans and animals. Hence, CO₂ generation and emissions must be minimized. Alternatively, finding ways to capture, store, and utilize carbon dioxide could solve this problem of global warming and health hazards. Hence, this review presents and proposes carbon dioxide capture, transportation, utilization, and storage (CCTUS) to generate energy for future development. This work shows CCTUS technology for mitigating CO₂ emissions. This review presents pre, post, and oxy-fuel combustion to capture and utilize carbon dioxide emissions to convert electrical energy, hydrogen, and water. The paper also presents less focused options (algae, biochar, nanotechnology, and so on) for capturing CO₂. This work also explores different transportation and storage options with their advantages and disadvantages. Additionally, the potential of CO₂ utilization for future fuels including methanol, dimethyl ether, formic acid, and biofuel has been shown. This review presents that CO₂ can be used in producing valuable products like chemicals, fuels, and biofuels, and this review emphasizes its role in creating economic incentives and enhancing the sustainability of CCTUS technologies. By integrating recent research findings and technological innovations, this paper provides a holistic view of CCTUS, underscoring its critical role in achieving long-term climate goals and industrial sustainability.

Keywords Carbon dioxide · Greenhouse gas emissions · Carbon dioxide capture · Transportation · Utilization · Storage · Sustainable energy solutions · Climate change mitigation

Introduction

Protection of the environment is crucial for human welfare because industrialization significantly increases air pollution. Coal-fired power plants, fossil-fueled vehicles, a solid waste incinerator, a cement plant, and value-added industries such as garments, textiles, fertilizer, pharmaceuticals, and agricultural industries including steel, paper, sugar, oil shale, pulp, and chemical industries are substantial contributors to air pollution (Chea et al. 2022; Sharma et al. 2024). These cause air

pollution by producing carbon dioxide (CO₂), nitrogen oxides (NO_x), sulfur oxides (SO_x), and soot particles (Hasan et al. 2020). Nitrogen oxides (NO_x), carbon monoxide (CO), sulfur dioxide (SO₂), hydrocarbons, and particulate matter are some of the gases that are found in the air, and depicts as the greenhouse gases (GHGs) emitted by industry. Heavy industries and cars combust fossil fuels. Therefore, the amount of CO₂ that these industries and vehicles release into the environment at an alarming rate raises the earth's average temperature. CO₂ emissions considerably affect global climate due to the continuous rise in atmospheric carbon dioxide concentration, leading to a marked increase in greenhouse gasses and the consequent greenhouse effect in this decade (Joarder et al. 2019). Figure 1 depicts countries with large populations, like India and China, emitting more CO₂ emissions, while countries like Australia, Brazil, Qatar, Indonesia, the Philippines, Mexico, Bangladesh, and Vietnam emitting less (Ghezelbash et al. 2023).

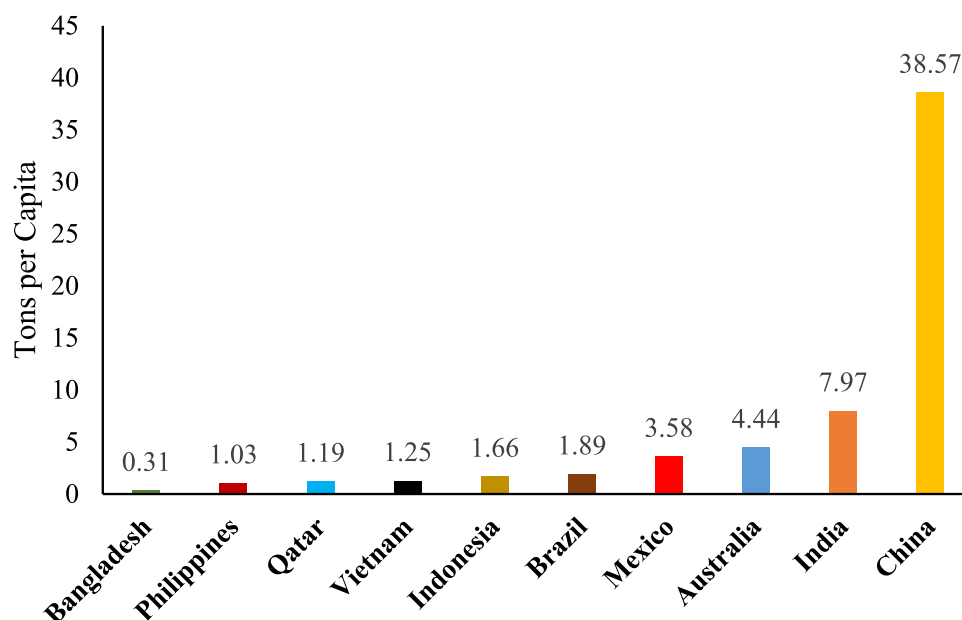
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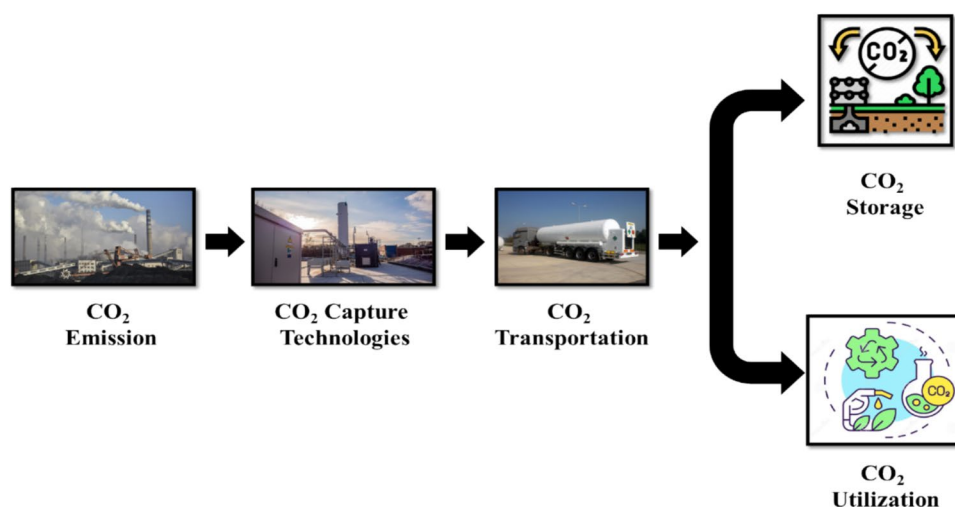
Fig. 1 Scenario of per capita carbon dioxide (CO₂) emissions for different countries (Adapted from Ghezelbash et al. 2023)



One of the most highly effective techniques is carbon dioxide capture, transportation, utilization, and storage (CCTUS). It encompasses the elimination of CO₂ emissions from industrial and energy-related sources, the transportation of CO₂ to storage sites or facilities for utilization, the conversion of CO₂ into valuable goods, and the sequestration of CO₂ in subterranean geological formations or alternative long-term storage solutions (Siyuan Chen et al. 2022; McLaughlin et al. 2023; Mikulčić et al. 2019). The efficacy of CCTUS is contingent upon a variety of factors, such as the form of combustion process used, the amount of CO₂, and how that CO₂ is separated, transported, or absorbed for reuse or storage (Esiri et al. 2024). Besides all these things, there should be some critical criteria in these separation processes, such as low operating costs, low environmental impact, no risk of accident, and low energy consumption (E. Liu et al. 2023). CCTUS technologies may enable industries to use cheap carbon-based feedstock at the same time meeting carbon dioxide discharge targets (Bui et al. 2018). Since CCTUS can readily transform waste carbon dioxide emissions into valuable resources like chemicals and fossil fuels while also lessening the impacts of climate change, it has gained attention and recognition on a global scale (Hasan et al. 2020; Tushar, Islam, Ahmmed, & Joarder, 2024). In 1996, the Sleipner project in Norway, which is among the most renowned and early CCTUS operations, commenced the injection of CO₂ into a saline aquifer located beneath the North Sea (Luo et al. 2023). Investment in CCTUS technology aimed at mitigating greenhouse gas emissions and comprehending climate change surged significantly between the 2000s and 2010s (Babakhani et al. 2022; Usman et al. 2022). With its emphasis on cutting greenhouse gas emissions and mitigating global warming to less than 1.5 °C, the 2015 Paris Agreement attracted fresh investment

and research in CCTUS technology (K. Zhao et al. 2023). Increased funding and research efforts have resulted in considerable breakthroughs in CCTUS technologies these past few years. The creation of more economical and effective carbon capture systems has been the main priority (Shang and Lv 2023). By drastically lowering the cost and amount of energy needed for CO₂ capture, these technologies can make it more practical for a greater variety of uses (Bahman et al. 2023). The purpose is to provide new uses for captured CO₂, such as the production of chemicals, fuel, building materials, and better recovery through oil and gas field injection (K. Zhao et al. 2023). Such applications have the potential to create new markets for CO₂ collection, increasing the viability of capturing and storing it economically (Koytsoumpa et al. 2018). The adverse effects of climate change could be lessened and emissions of greenhouse gases drastically reduced with the help of recent developments in CCTUS technologies. A 9-point rating system called the Technical Readiness Level (TRL) is utilized to gauge the technological maturity and advancement of CCTUS technology (Dziejarski et al. 2023). Faster commercial adoption of CCTUS technology is essential for reducing CO₂ emissions over the long run in a variety of industries (Loria & Bright 2021). Figure 2 illustrates the sequential stages beginning with CO₂ emissions from industrial and energy sources, followed by CO₂ capture technologies for separation and collection. The captured CO₂ is then transported via pipelines, trucks, or other methods to designated facilities, where it is either stored in long-term geological formations or utilized for value-added applications such as manufacturing building materials, chemicals, and fuels. This integrated approach highlights the multifaceted strategy for reducing atmospheric CO₂ concentrations and mitigating climate change.

Fig. 2 Carbon dioxide capture, transportation, utilization, and storage (CCTUS) technology



Despite significant advancements, critical challenges continue to hinder the widespread adoption of CCTUS technologies. Davoodi et al. (2023) provided a detailed review of technological progress in carbon capture and sequestration (CCS), highlighting its potential to mitigate climate change. However, their analysis did not fully address the economic and technical barriers that limit the scalability of these technologies (Davoodi et al. 2023). Similarly, Gowd et al. (2023) examined the economic feasibility of CCS and emphasized the need for policy mechanisms such as carbon pricing to drive adoption. Nonetheless, their review lacked a comprehensive discussion on how emerging technologies could reduce costs and improve efficiency (Gowd et al. 2023). Bhavsar et al. (2023) focused on integrating CO₂ utilization within industrial processes, yet they did not explore cutting-edge solutions such as machine learning, advanced membranes, or nanoparticle-based systems that could revolutionize carbon capture (Bhavsar et al. 2023). Davoodi et al. (2024) investigated CO₂ sequestration through enhanced oil recovery (EOR), which offers the dual benefits of emission reduction and resource recovery, but their work overlooked alternative storage methods like mineral carbonation and non-geologic options (Davoodi et al. 2024).

Building on these prior studies, it is evident that significant gaps remain in literature. First, there is limited exploration of emerging technologies, including algae-based CO₂ capture, direct air capture, and biochar utilization, which show potential for scalability and commercialization. Second, the integration of economic and policy frameworks with technological advancements is underexplored, leaving challenges related to funding, incentives, and market dynamics unaddressed. Lastly, a comprehensive review that synthesizes advancements across CO₂ capture, transportation, storage, and utilization into a unified framework is lacking. Addressing these gaps is essential for accelerating the adoption of CCTUS

technologies and realizing their full potential in combating climate change.

This study aims to bridge these gaps by presenting a comprehensive review of CCTUS technologies, with a focus on emerging innovations and their practical applications. Unlike previous reviews, this work provides a detailed comparison of pre-combustion, post-combustion, and oxy-fuel combustion methods, evaluating their efficiency, energy consumption, and cost-effectiveness. Additionally, CO₂ transport, storage, and utilization approaches have been outlined as well. The paper also delves into innovative storage solutions, including geological and mineral carbonation methods, offering insights into their long-term viability and environmental impact. This review uniquely integrates the potential of CO₂ utilization in producing economically sound products such as chemicals, fuels, and biofuels, emphasizing the dual benefits of emission reduction and resource recovery. This study enhances the understanding of CCTUS technologies by integrating a comprehensive literature analysis with the most recent research discoveries and technological advancements. Additionally, it offers a strategic framework for using these technologies to address global climate change. The comprehensive approach emphasizes the crucial importance of CCTUS in attaining sustainable industrial practices and long-term environmental objectives. Lastly, the primary novelty and main contributions of this study could be summarized in the following:

- Recent carbon capture technologies like pre, post, and oxy-fuel combustion carbon captures have been discussed with their energy consumption, efficiency, costs, and so on. The opportunities and challenges have been discussed as well.
- The article has also emphasized the less carbon dioxide capture processes like direct air capture and CO₂ capture by algae, nanotechnology, biochar, and adsorbents.

- The captured CO₂ transportation, storage, and utilization processes have been discussed in detail as well.
- Moreover, most of the review articles discussed CCU/CCS technologies. But this article emphasizes CCTUS technologies, which is the main attractive part of this literature. This paper also shows the comparison among the different studies of each section.

Carbon dioxide capture technologies

Carbon capture frameworks include pre-combustion (CC), post-combustion (CC), and oxy-fuel combustion (OFC) capture, etc. (L. Zhao et al. 2008). Syngas, or synthesis gas, is created in Pre-CC when the fuel, air, oxygen, and steam combine. The resulting carbon monoxide (CO) subsequently interacts with steam in a catalytic shift converter to generate carbon dioxide (CO₂) and more hydrogen. The procedure of capturing CO₂ from the flue gas produced following the combustion of fossil fuels is referred to as post-combustion carbon capture (post-CC). There are a number of techniques for collecting CO₂ after combustion, including membranes, adsorption, absorption, and cryogenic technologies (Williams et al. 2012). In OFC, fossil fuels are combusted in an oxygen-saturated environment, and they predominantly produce CO₂ and steam, which facilitate the CO₂ capture process. CO₂ is present in very high concentrations in flue gas during oxyfuel burning. The various technologies and procedures for CO₂ separation are shown in Fig. 3 that categorizes carbon capture into three primary approaches: post-combustion, pre-combustion, and oxy-fuel combustion. Each method utilizes a range of techniques, including absorption, adsorption, cryogenic distillation, membrane separation, gas hydrates, and chemical looping, for efficient CO₂ separation and collection. The connections illustrate the flexibility and applicability of each technique across the different capture methods, highlighting their role in optimizing CO₂ capture efficiency (Kenarsari et al. 2013).

Fig. 3 Diverse techniques for capturing carbon dioxide (CO₂) emissions (Adapted from Bahman et al. 2023)

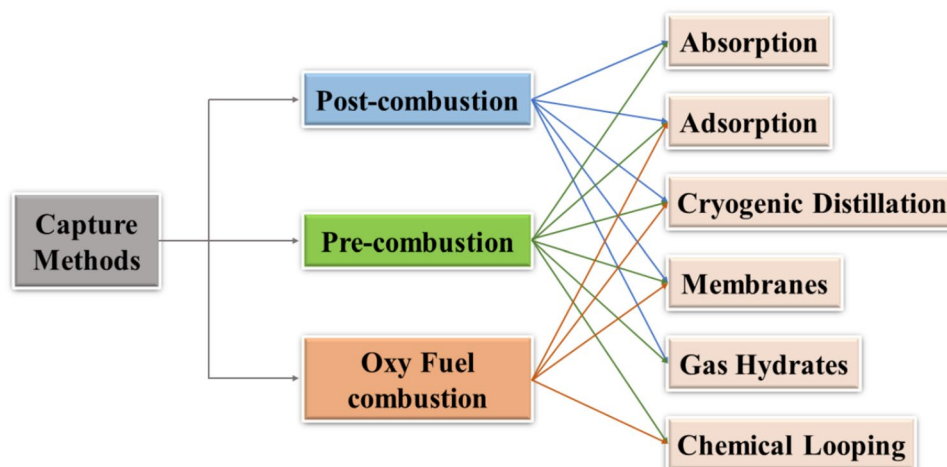


Table 1 compares key parameters across different CO₂ capture methods, including CO₂ concentration, combustion medium, equipment size, operating temperature and pressure, energy consumption per ton of CO₂ captured, removal efficiency, cost per ton of CO₂ captured, and technical readiness level (TRL). The data provides a comprehensive overview of the performance and feasibility of these technologies in mitigating CO₂ emissions.

Pre-combustion capture

Synthesis gas treatment (SGT), which is made from fossil fuel and mostly consists of H₂ and CO with a trace of CO₂, is a step in the pre-CC process (Figueroa et al. 2008). In this process, combustion starts with the generation of syngas, the extraction of CO₂, and finally H₂ combustion (Steenneveldt et al. 2006). In water gas shift reactions, the temperature reduction and addition of steam increase the conversion of CO into CO₂ and the enhancement of H₂ generation. After that, CO₂ is separated from the H₂ and CO₂-bearing streams, and the separated carbon dioxide is sent to the compression unit. For electricity production, the separated pure H₂ is sent as an input to a combined cycle (CC) (Fig. 4) (Mondal et al. 2012). The separation process mainly uses physical solvents like selexol and rectisol, which are less expensive. The CO₂ is dispersed at a higher pressure (HP) and then released due to a decrease in pressure. The production of solvents does not require the use of heat, and CO₂ may be released with ease at pressures higher than atmospheric. Research on various pre-combustion methods is not new. There have been numerous research proposals and strategies in the last decade as shown in Table 2.

Subraveti et al. (2019) exemplified that novel pressure-swing adsorption (PSA) adsorbent is stable at high pressure (HP) and was proven through testing over more than 11,650 cycles. The cycles were assessed for their ability

Table 1 Efficiency of CO₂ removal and energy consumption across various capture technologies

CO ₂ capture technology	CO ₂ concentration (vol %)	Combustion medium	Equipment size	Temperature	Pressure	Energy consumption for CO ₂ capture (GJ/tCO ₂)	CO ₂ removal efficiency (% vol.)	Cost (USD t ⁻¹ of CO ₂)	TRL	References
Pre-combustion	15–40	Steam/air	Medium size	Low/high (700–1300 °C)	High (20–30 atm)	3.35	90	34–63	3–9	Dinca et al. (2018a, b); Dinca et al. (2018a, b); Dziejarski et al. (2023); Hong and Technology (2022); Kato et al. (2006); Raganati and Ammendola (2024); Rubin et al. (2015); Rubin et al. (2015)
Post-combustion	4–20	Air	Large size	Low/high (400–800 °C)	Low (0.15–1 atm)	4.14	90	46–74	2–9	Dinca et al. (2018a, b); Dziejarski et al. (2023); El Nasr et al. (2015); Hong & Technology (2022); Raganati et al. (2024); Rubin (2015); Rubin et al. (2015)
Oxy-fuel combustion	75–80	High-purity oxygen	Small size	High (up to 2000 °C)	Low (~1 bar)	4.05	> 90	52	4–7	Hong & Technology (2022); Li, Duan, Yang, Zhao, & Flame (2020); Loria & Bright (2021); Raganati et al. (2024); Rubin (2015); Rubin et al. (2015); Senior et al. (2013) Vega et al. (2019)

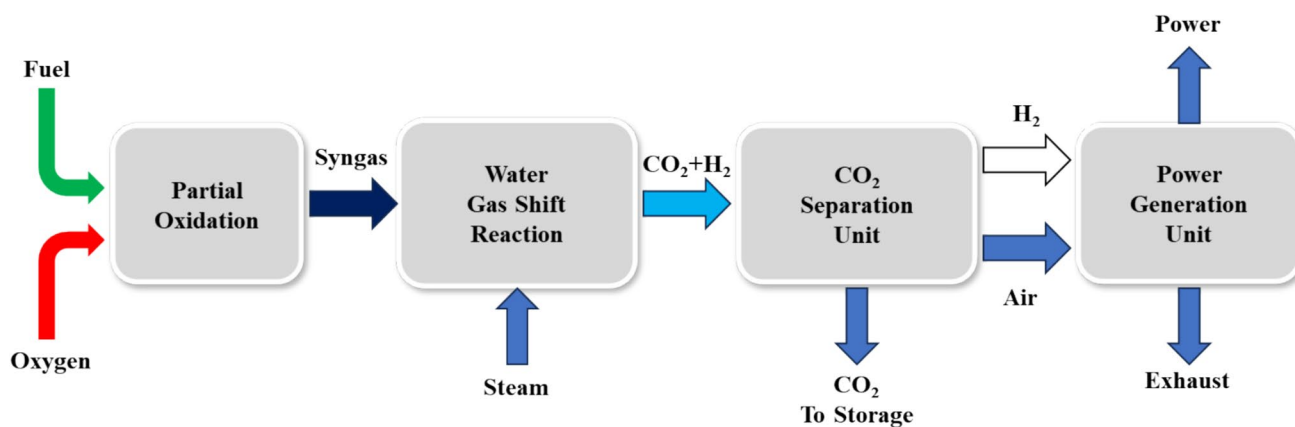


Fig. 4 Carbon dioxide (CO₂) emissions capture by using pre-combustion technique (Adapted from Blomen et al. 2009; Damen et al. 2006; Figueroa et al. 2008)

to achieve high CO₂ purity (> 95%) and recovery (> 90%) using multi-objective optimization (Subraveti et al. 2019). Sohaib et al. (2020) study various membrane contactor modules with various fiber counts which were compared. As the quantity of fibers escalated from 1 to 10, the efficiency of CO₂ separation rose by 31%. 1-Ethyl-3-methylimidazolium glycinate [C₂mim] [Gly], tetramethylammonium glycinate [N1111] [Gly], 1-hexyl-3-methylimidazolium, and 1-butyl-3-methylimidazolium glycinate [C4mim] [Gly] were four amino acid-based ionic liquids (ILs) utilized as absorbents in Sohaib et al. (2021). Hospital-Benito et al. (2022) investigated four distinct regeneration pressures—0.1, 0.25, 0.5, and 1 bar—as well as a range of absorption and regeneration temperatures (between 30 °C and 120 °C) to increase the efficiency and lower the capital and operational expenses of a pre-combustion CO₂ capture (Hospital-Benito et al. 2022). A novel porous slurry made of ZIF-8 and isoparaffin C16 enhances the performance and selectivity of CO₂ absorption in place of more conventional physical absorbents, and the slurry shows significantly higher selectivity over H₂ (17.4–205, which could be one time higher than DEPG and three times higher than PC), as well as a much higher CO₂ sorption capacity (Kim et al. 2023).

Post-combustion capture

At atmospheric pressure, thermal power plants (TPP) generate flue gases containing CO₂ by utilizing air during combustion. The post-combustion procedure involves removing CO₂ from waste gas streams once the carbon supply has been burned to produce CO₂ or digested wastewater sludge (Fig. 5). In a number of industries, it may be utilized to remove CO₂, including power plants (PPs), cement, ethylene oxide production, iron, fuel production, and steel production, as well as biogas sweetening (Alexanda Petrovic & Masoudi Soltani 2019). Flue gases

typically contain less than 15% CO₂, so capturing this low concentration of CO₂ requires little thermodynamic driving power (Nandakishora et al. 2021). As the concentration of CO₂ is low in flue gases, PPs must handle enormous amounts of gas (Fernando Vega et al. 2017). As a result, the PPs requires large equipment, which has an elevated capital expenditure. The study of different post-combustion techniques is not distinctive (Feron et al. 2017). Over the past 9 years, there have been many different study plans and ideas. Table 3, from a variety of literature sources, displays the most recent advancements.

Sher et al. (2020) compares three biomass categories for CO₂ adsorption capabilities and the thermal gravimetric analyzer (TGA) examined the adsorption isotherms and found that at 25 °C, CO₂ could be adsorbed within the parameters of 2.53–3.92 mmol/g (over N₂ balance) (Sher et al. 2020). Park et al. (2021) investigated that the efficient carbon adsorbent can hold 3 mmol g⁻¹ of CO₂ at 1 bar. Selectivity of 534 for H₂/CO₂ (60:40) and 22.6 for N₂/CO₂ (50:50) were displayed by the most effective adsorbent (Park et al. 2021). A power plant is pumped into the aqueous post-combustion CO₂ capture (APCC) that features a 72 kt/year CO₂ collection capability, in order to develop and evaluate a novel onshore technique for sequestering CO₂ from flue gases by means of water as a physical solvent (Malekli & Aslani 2022). Gao et al. (2023) investigate the effects of polyamines' molecular structures regarding their reaction kinetics, CO₂ absorption heat, and CO₂ cyclic capacity and decreases CO₂ absorption heat by 56.3% in comparison to a 5 M aqueous solution of MEA, indicating its potential as an absorbent (Gao et al. 2023).

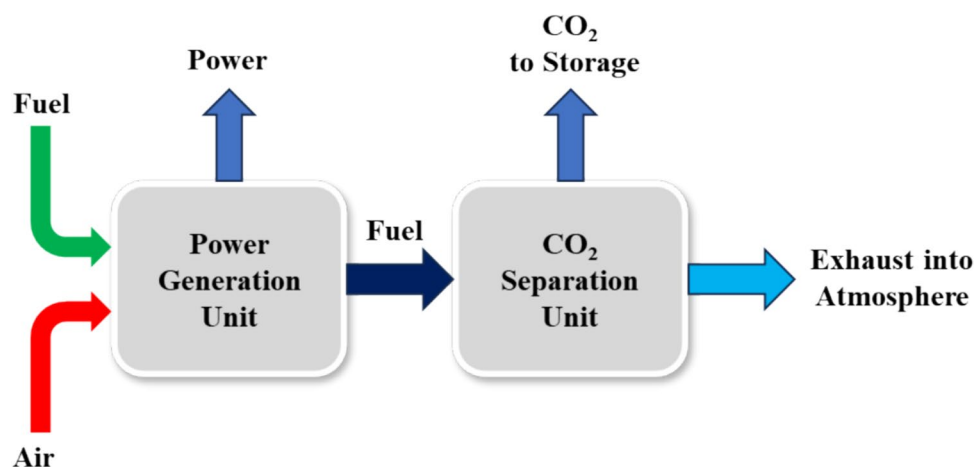
Oxy-fuel combustion capture

CO₂ collection is facilitated by oxygen-fuel combustion, which burns using approximately 95% pure oxygen and

Table 2 An overview of some of the most significant findings on pre-combustion carbon capture (pre-CCC)

Objective	Methodology	Findings	Strength	Weakness	Year	Ref
To assess the viability of pre-combustion capturing CO₂	It was demonstrated that novel pressure-swing adsorption (PSA) adsorbent is stable at high pressure and was proven through testing over more than 11,650 cycles	The study utilizing a 10-step cycle with three pressure equalization phases can achieve an energy consumption as low as 95.7 kWh per tonne of CO₂ captured, with a productivity rate of 3.3 mol/m³ CO₂ of adsorbent per second	The cycles were evaluated for their capacity to produce high CO₂ purity (> 95%) and recovery (> 90%) using multi-objective optimization	The potential difficulties in putting the suggested PSA cycles for pre-combustion CO₂ capture into practice are not covered here	2019	Subraveti et al. (2019)
To research a pre-CCC procedure employing an ionic liquid absorbent in a membrane contactor setup (MCS)	In this study, various membrane contactor modules with various fiber counts were compared	The addition of more fibers has significantly increased the process of CO₂ absorption. When the number of fibers was increased from 1 to 10, the efficiency of CO₂ separation rose by 31%	It provides a thorough analysis to comprehend CO₂ mass transfer in a membrane contactor configuration, taking into account both axial and radial molecular diffusion	The study suggests that future work should focus on studying the pre-CCC with MCS using a 2D mass transport model	2020	Sohaib et al. (2020)
In formulating a mathematical model for hollow fiber membrane contactors (HFMC) for a CO₂ absorption process that is non-isothermal	1-Ethyl-3-methylimidazolium glycinate [C₂mim] [Gly], tetramethylammonium glycinate [N1111] [Gly], 1-hexyl-3-methylimidazolium, and 1-butyl-3-methylimidazolium glycinate [C4mim] [Gly] were four amino acid-based ionic liquids (ILs) used as absorbents	Verification of the substantial impact of absorbent concentration and the negligible effect of process temperature on the CO₂ border flow, ILs' ability to separate, and the kinetics of the interaction between CO₂ and ILs	Validation of the developed model through comparison with experimental data, confirming the identical trends and validating the model	This study has to be extended for tough industrial working conditions and further modeling and experimental work is required to make it a viable technology	2021	Sohaib et al. (2021)
To increase the efficiency and lower the capital and operational expenses of a pre-CCC	An ionic liquid-based [P2228] [CNPy] chemical absorption-based pre-CCC technique was evaluated	Investigations were conducted using four different regenerating pressures—0.1, 0.25, 0.5, and 1 bar—as well as a range of absorption and regeneration temperatures (between 30 °C and 120 °C)	The study lowers CO₂ emissions from power generation and promotes the use of hydrogen as a feedstock, fuel, or energy carrier	More life cycle assessments may be needed to fully assess these IL-based systems' sustainability	2022	Hospital-Benito et al. (2022)
A novel porous slurry made of ZIF-8 and isoparaffin C16 to improve the selectivity and capacity of CO₂ absorption	Experimental research has been done on the characteristics and sorption behaviors of this slurry, and 20 weight percent of ZIF-8 is the ideal content	By vacuuming for 5 min without heating, the 98% sorption capacity is regained, demonstrating that pressure swing is the only way to complete the simple sorption–desorption cycle	The slurry exhibits a much greater CO₂ sorption capacity in addition to a significantly higher selectivity over H₂ (17.4–205, potentially one time higher than DEPG and three times higher than PC)	Additional research on ZIF-8/isoparaffin C16 slurry is required to achieve reduced energy costs and increased efficiency	2023	Kim et al. (2023)

Fig. 5 Carbon dioxide (CO₂) emissions capture by using post-combustion technique (Adapted from Wang et al. 2017)



recovers flue gas (RFG) instead of air. The exhaust emits flue gases that comprise water vapor and higher concentrations of CO₂, and CO₂ capture units can produce highly concentrated CO₂. Instead of ambient air, almost pure oxygen is utilized in this process for burning, resulting in mostly H₂O and CO₂, that can be readily detached using condensing water (Fig. 6) (Blomen et al. 2009; Damen et al. 2006; Figueroa et al. 2008). This lowers the exhaust gas' nitrogen content, which affects the subsequent separation process. There have been numerous distinct study strategies and concepts during the previous decade. Table 4 outlines the objectives, methodologies, findings, strengths, and weaknesses of selected studies focused on OFC-based carbon capture. This compilation highlights the advancements, challenges, and research contributions in optimizing OFC technology for CO₂ mitigation.

Wu et al. (2020) tested a 50-kg/h platform for purifying and compressing CO₂, demonstrating high-concentration CO₂ collection and viability (H.-b. Wu et al. 2020). Three case studies are compared in Perpiñán et al. (2021): case 0 is a standard ironmaking process; case 1 is a power-to-methane integration with water electrolysis; and case 2 is a power-to-methane integration with coke oven gas. This study aims to reduce CO₂ emissions by 8% and coal fuel usage by 12.8% in the first deployment. The second deployment requires 4.9 MJ/kg CO₂. Huang et al. (2022) studied that oxy-fuel combustion captures CO₂ in flue gas maintaining a 2% volume percentage and 98.2% desorption efficiency at 0.01 MPa desorption pressure, making it a cost-effective strategy (X. Huang et al. 2022). Fei et al. (2023) used a LiBr-H₂O double-effect absorption refrigeration cycle with an oxygen separator membrane (OSM), an oxygen-fuel cycle, and a CO₂ liquefaction cycle to get 95% energy efficiency and 38% exergetic efficiency (Fei et al. 2023).

However, Table 5 highlights the specific opportunities offered by pre-combustion, post-combustion, and oxy-fuel combustion CO₂ capture technologies, the associated

challenges, and their respective development stages. It provides a comprehensive overview of their current technological readiness and future potential, supported by relevant references.

Less focused carbon dioxide capture processes

Direct air capture

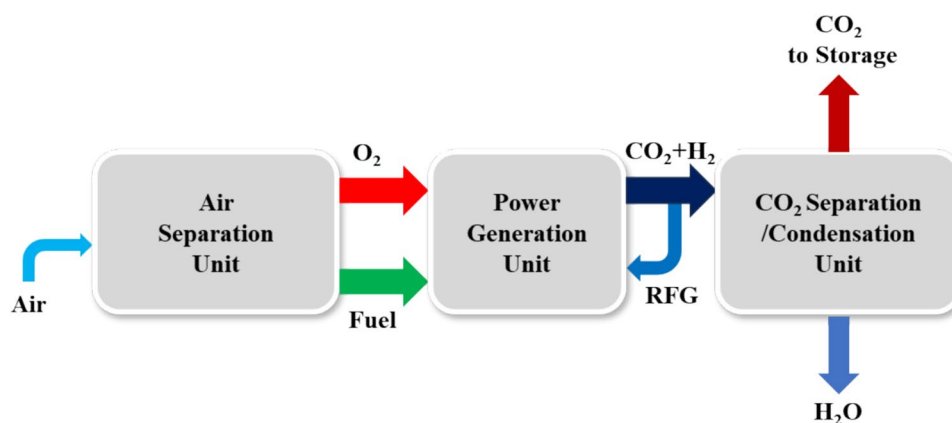
Direct air capture (DAC) is a prominent method that extracts CO₂ directly from the surrounding atmosphere, in addition to other technologies that lower emissions. A potential synthetic carbon sink is provided by this technology to reduce industrial emissions that point source solutions were unable to adequately collect. DAC plant site can be selected separately from emission source location, increasing the flexibility of CO₂ transportation and possibly lowering infrastructure expenses. Waste heat from DAC plants close to power plants can be used to fulfill high energy demands (McQueen et al. 2021). Capturing atmospheric CO₂, regardless of release, can offset emissions from sectors like aviation and remote or decentralized industrial processes that are harder to decarbonize (Aldaco et al. 2019). Figure 7 illustrates the increasing demand for DAC technology over time, with the transport sector showing the highest demand growth, followed by the industry sector, while the power (gas) sector remains comparatively lower. The data highlights the pivotal role of DAC in decarbonizing key sectors to achieve net-zero emissions targets.

However, this procedure was primarily introduced by Lackner in 1999. With its early demonstration stage (TRL 7) setting the groundwork for its potential future effect, this ground-breaking technology is gaining attention. Recently, the potential of DAC to reduce CO₂ emissions from all sources has drawn emphasis, not just large point sources.

Table 3 Some of the most important experiments on post-combustion carbon capture (post-CCC)

Objective	Methodology	Findings	Strength	Weakness	Year	Ref
To produce adsorbents or activated carbons for post-CCC	This study compares three biomass categories for CO₂ adsorption capabilities at 25 °C, focusing on soft, intermediate woody, and hard woody biomasses	The thermal gravimetric analyzer (TGA) examined the adsorption isotherms and found that at 25 °C, CO₂ could be adsorbed within the parameters of 2.53–3.92 mmol/g (over N₂ balance)	To create activated carbons (ACs) for post-CCC from biomass waste, the study suggests an innovative and environmentally friendly method that can lower expenses and risks to the environment	The performance of the ACs in an actual post-combustion CO₂ collection system is not experimentally validated; instead, the research solely presents the TGA	2020	Sher et al. (2020)
To create selective and effective nanoporous carbon adsorbents for post- and pre-combustion CO₂ collection	A variety of methods are employed to examine the MIP, MIP composite, and ACs' structure, morphology, porosity, and surface chemistry. FTIR, TGA, Raman spectroscopy, XRD, XPS, SEM, and N₂ adsorption–desorption assays were among these methods	The efficient carbon adsorbent can hold 3 mmol g^{−1} of CO₂ at 1 bar. Selectivity of 534 for H₂/CO₂ (60:40) and 22.6 for N₂/CO₂ (50:50) were displayed by the most effective adsorbent	The study describes a simple one-pot method for creating molecularly imprinted polymers (MIPs) and MIP composites with activated carbon that are extremely porous and oxygen-rich carbon adsorbents	There are no experimental data on the performance of the nanoporous carbon adsorbents provided in this study; it just reports on their manufacture and characterization	2021	Park et al. (2021)
To create and evaluate a new onshore method for CO₂ capture from flue gas called aqueous post-combustion CO₂ capture (APCC)	An aqueous post-combustion CO₂ capture (APCC) technique with excellent energy efficiency was developed here in Aspen Plus™, utilizing water acts as a physical solvent for the extraction of CO₂ from flue gas	Flue gas resulting from a power plant is pumped into the APCC with 72 kt/year CO₂ collection capability and managed to capture nearly 97.4% of the CO₂ at a ratio of power production to consumption of 49.5% (22.74 MWh/ton CO₂)	With the use of an organic Rankine cycle, a hydro-pump, and a unique design, it suggests an aqueous post-combustion CO₂ capture technique that may produce large amounts of electricity while consuming little energy	The study notes that high pressures and low temperatures enhance CO₂ solubility in water, resulting in significant energy consumption and equipment expenses for compression, cooling, and pumping	2022	Malekli & Aslani (2022)
To investigate the effects of polyamines' molecular structures regarding their CO₂ absorption heat, reaction rate, and CO₂ cyclic capacity	With the use of a high-precision microcalorimeter, it measures sensible and absorbed heat of CO₂ directly, yielding more precise and trustworthy results than indirect approaches relying on VLE data	The efficient phase change solvent system of 2MPRZ + triethylene glycol monobutyl ether (TGBE) + H₂O decreases CO₂ absorption heat by 56.3% in comparison to a 5-M MEA aqueous solution	Using 2MPRZ as a foundation, it creates a phase change absorbent system with promising potential to lower regeneration energy consumption and increase CO₂ capture efficiency	The paper used a microcalorimetric method to directly measure the CO₂ absorption heat, it may have some errors and uncertainties due to the experimental conditions and the accuracy of the instrument	2023	Gao et al. (2023)

Fig. 6 Carbon dioxide (CO₂) emissions capture by using oxy-fuel combustion (Adapted from Blomen et al. 2009; Damen et al. 2006; Figueroa et al. 2008)



DAC currently operates two commercial routes: (i) liquid solvent-based method (LSBM), (ii) solid sorbent-based method (SSBM) (Sanz-Pérez et al. 2016).

LSBM absorbs atmospheric CO₂ using a liquid solution. When CO₂ interacts with a strong alkaline liquid calcium hydroxide (Ca(OH)₂), calcium carbonate (CaCO₃) is formed; heating CaCO₃ releases concentrated CO₂, which is then converted to calcium oxide (CaO). Calcium oxide (CaO) is combined with water to generate calcium hydroxide (Ca(OH)₂) which can be used again (Sanz-Pérez et al. 2016). A solid sorbent, such as carbon nanotubes, silica materials, metal–organic frameworks (MOFs), zeolites, activated carbons, silica materials, amine-modified materials, and carbon molecular sieves, is where CO₂ is adsorbed in SSBM (McQueen et al. 2021). The process incurs a substantial energy penalty of around 15 GJ per tonne of CO₂ for the regeneration of Ca(OH)₂, which will raise operational costs even though LSBM allows for adequate CO₂ separation (Bacocchi et al. 2006). On the other hand, because SSBM regenerates solid sorbents at reduced temperatures (ranging from 80 to 100 °C), it uses less energy (up to 1 GJ/t CO₂), lowering CO₂ capture costs at the DAC plant (Fasihi et al. 2019; Sinha & Realff 2019). DAC systems based on passive CO₂ carbonation of porous Ca(OH)₂ plates cost USD 140–340/t CO₂, according to a recent study (Abanades, Criado, & Fernández, 2020). The expense of DAC is diminished through the use of economical polyvinyl chloride (PVC) gas–liquid contactors (Keith et al. 2018). Compared to DAC systems utilizing LSBM, the cost of DAC using SSBM was lower, ranging from USD 86/t CO₂ to USD 221/t CO₂ (Sinha & Realff 2019). Aqueous solutions of potassium hydroxide (KOH) and sodium hydroxide (NaOH) are two supplementary liquid solvents with a significant ability to bind CO₂, which have been used as substitutes to lessen the high expense (Fasihi et al. 2019). Recently, DAC has been widely displayed by a number of companies. Carbon Engineering's DAC facility, inaugurated in 2015 in British Columbia, Canada, sequesters around one tonne of CO₂ per day utilising a potassium hydroxide solution as a sorbent

(Keith et al. 2018). Since 2017, Climeworks' DAC plant in Hinwil, Switzerland, has captured 900 tons of CO₂ per year (Keith et al. 2018). The combined capture rate of 19 DAC plants in the USA, Canada, and Europe as of November 2021 is more than 0.01 Mt CO₂/year. By 2050, this new carbon removal technique might absorb about 980 Mt CO₂ yearly, allowing for storage and use in sectors including synthetic fuels and food processing. The most extensive DAC facility in Iceland, which sequesters 4000t CO₂/year in basalt rock formations, is operated by Climeworks and CarbFix. The first 1 Mt CO₂/year DAC plant is being built in the USA and is expected to start operations in 2024 (McLaughlin et al. 2023). Significant energy requirements for capture and sorbent regeneration, coupled with substantial initial investment, operational costs, and sorbent maintenance expenditures, are major obstacles for DAC (Fuss et al. 2018). By lowering CO₂ emissions, DAC can slow down climate change. However, it may also restrict photosynthetic activity in some places, which could have an impact on natural carbon capture systems. Because future technical developments may reduce costs, early-stage development strategies like DAC are worth taking into consideration. Although it requires overcoming resource and financial hurdles, DAC, an alternative negative emission technology capture method, provides significant benefits compared to bioenergy with carbon capture and storage due to lower land, water, and energy requirements (Azarabadi and Lackner 2020).

Carbon dioxide capture by algae

One hundred tonnes of microalgae (MA) can dispel 183 tonnes of CO₂. As seen in Fig. 8, algae growth depends on beneficial factors like pH, water, light, temperature, nutrients, and CO₂ (Anwar et al. 2018). In comparison to terrestrial plants, MA have a simpler structure, higher CO₂ fixation efficiency, and a quicker development rate. It is a more efficient and cost-effective process since wastewater is used as the raw material (Ghosh & Kiran 2017). A number of species, notably *Chlorella* sp., *Botryococcus* b., *Chlorella*

Table 4 A summary of some key studies on oxy-fuel combustion (OFC) carbon capture

Objective	Methodology	Findings	Strength	Weakness	Year	Ref
• To develop a CO₂ compression and purification technology for OFC that can streamline the (CCS) process	• This investigation tested a 50 kg/h platform for purifying and compressing CO₂, demonstrating high-concentration CO₂ collection and viability	• The system's average desulfurization rate and denitration efficiencies exceeded 98%, and no change in the n(SO₂):n(NO_x) ratio affected sulfur and nitrate purification	• The best operating conditions and control strategies for CO₂ liquefaction, energy usage, and pollutant removal are examined	• There isn't any water vapor in the gas distribution system but there is some water present throughout the compression process (led to compressor corrosion)	2020	H.-b. Wu et al. (2020)
• To assess the CO₂ emission reductions of the suggested integrations compared to a conventional process of a steel industry	• Three case studies are compared in the paper: case 0 is a standard ironmaking process; case 1 is a power-to-methane integration with water electrolysis; and case 2 is a power-to-methane integration with coke oven gas	• This study aiming to reduce CO₂ emissions by 8% and coal fuel usage by 12.8% in the first deployment. The second deployment requires 4.9 MJ/kg CO₂	• This article addresses the issue of reducing CO₂ emissions in the iron and steel sector in a compelling and relevant manner	• The perfect gases, adiabatic reactions, and clean streams of O₂, H₂, and CH₄ assumed in the investigation could not accurately represent the actual processes or their performance	2021	Lau et al. (2021)
• To design and simulate the method of CO₂ separation in flue gas from the OFC power plants	• The authors examine the simulation outcomes, including the energy usage, product purity, solvent circulation, and CO₂ recovery. Moreover, they contrast the OFC method with others	• OFC captures CO₂ in flue gas maintaining a 2% volume percentage and 98.2% desorption efficiency at 0.01 MPa desorption pressure, making it a cost-effective strategy	• It proposes an innovative design of CO₂ capture process using ionic liquid [emim][Tf2N] for flue gas from oxy-fuel combustion plant	• The air separation system is integral to the OFC process. Its energy consumption is not taken into account in this research	2022	X. Huang et al. (2022)
• To suggest a zero-emission power generating and cooling system for vessels, integrating different cycles	• The suggested system's technique, which comprises of three primary cycles: CO₂ liquefaction cycle, SCO₂-based oxy-fuel cycle, and oxygen separator membrane	• This study used a LiBr-H₂O double-effect absorption refrigeration cycle with three primary cycles to get 95% energy efficiency and 38% exergetic efficiency	• Multi-objective optimization increased exergetic efficiency, cost per exergy unit, and environmental impact per energy unit by up to 18% and 13%, respectively	• The ionic conductivity and oxygen flow of the tubular BSCF membrane used here are fixed, which may not accurately represent the membrane's true performance	2023	Fei et al. (2023)

Table 5 The opportunities and challenges in pre, post, and oxy-fuel combustion capture

Capture technologies	Opportunities	Challenges	Development stage	References
Pre-combustion	• Fewer capital expenditures and equipment requirements • Pre-combustion techniques may be 38–45% and 21–24% more cost-effective than post-combustion and oxy-fuel procedures, respectively • Making use of physical solvents, which are inexpensive and regenerate with little energy, that are employed in CO₂ separation • Possibility of easily switching between electricity and H₂ production, allowing for more output flexibility • Synthesis gas is an additional fuel source for the turbine cycle • H₂ can be used in transportation, fuel cells, and as a building block for high-value compounds production after further purification • This approach utilizes a lot less water than post-combustion capture	• Energy loss is greater than with post-combustion capture methods due to the energy needed for air separation and reformation • Throughout the production process, improvements in energy recovery efficiency must be pursued • Requires the placement of a chemical plant before the turbine • Retrofitting existing buildings makes the setup more expensive and difficult, which makes it impossible to commercialize • Operation under high pressure • Additional plant shutdowns are frequently the outcome of complicated chemical processes • Non-gaseous feedstocks (need for a cleansed gas stream and potential for expensive scrubbing to reduce high NO_x emissions) • The gasification stage is improved • The temperature at which H₂-rich gas causes heat transfer issues • Cooling of the flue gas is necessary to extract CO₂ from flue gas • Necessitates a sizable supporting infrastructure (such as an air separation unit) • A reduction in efficiency in the water–gas shift portion • To prevent any loss in solvent quality, the regeneration temperature for the absorption method should be lower than it is now. Ionic liquids, which have very little volatility, are being used to alleviate this issue	• Commercially available • Some large-scale projects	Elhenawy et al. (2020); Lau et al. (2021); Mondal et al. (2012); Omodolor et al. (2020); Osman et al. (2020); Osman et al. (2021); Portillo et al. (2019); X. Wang & Song (2020); K. Zhao et al. (2023)

Table 5 (continued)

Capture technologies	Opportunities	Challenges	Development stage	References
Post-combustion	• Relevant for both recently constructed and existing coal-fired power plants (utilizing current technology) • A lot of research is done to improve sorbents and capture equipment • Retrofitting current power plant designs is a practical choice • The post combustion process has greater thermal efficiency than pre-combustion for converting fuel to the electricity • The amount of CO₂ released is significantly low (80–95% of the CO₂ is recovered via adsorption, and 90–98% is by absorption). Additional NO_x and SO_x elimination • Hybrid approaches, including membrane-pressure swing adsorption, are employed to enhance the efficacy of CO₂ sequestration • Future pulverized coal systems with improved efficiency will produce more productive plants with lower CO₂ emissions	• Since there is vast amount of gas to handle, the low concentration of CO₂ at ambient pressure (usually 7–14%) necessitates big process equipment sizes and high prices • The flue gases' high temperature and comparatively low CO₂ partial pressure present a design difficulty • The absorption process for capturing CO₂ at low concentrations necessitates the use of strong chemical solvents, and the regeneration of these solvents to release CO₂ requires a substantial amount of energy • The amine technologies in the absorption method leads to a decrease in efficiency of 11% and a net power production decline of over 30% • The corrosive nature of amines, the substantial energy demands for regeneration, and deterioration are the primary factors contributing to solvent loss and evaporation during the absorption process • The MEA-based absorption approach is associated with high operating and capital costs. Hence, certain projects which used that technology has been put on hold • The reduced concentration of CO₂ in the flue gas yields a significant rise of approximately 60–70% in the additional cost of electricity production for new infrastructure and a much higher increase of around 220–250% for retrofitting • Traces of pollutants (NO_x and SO_x) have a significant impact on the adsorption technique • Steam extraction lowers the low-pressure turbine's capacity and efficiency by reducing the flow to it • For the adsorption separation process, a high-pressure drop is required • High capture levels need high performance, water consumption, and circulation volume	• Commercially available • Widely used at large-scale facilities	Bhattacharyya & Miller, (2017); Elhenawy et al. (2020); Lau et al. (2021); Mondal et al. (2012); Olajire (2010); Omodolor et al. (2020); Osman et al. (2021); Portillo et al. (2019); K. Zhao et al. (2023)

Table 5 (continued)

Capture technologies	Opportunities	Challenges	Development stage	References
Oxy fuel combustion	• It is applicable to both recently developed and existing power plants (using current technology) • CO₂ recovery is between 90 and 98% • Various fuels, including biomass and municipal solid waste, can be used, leading to the construction of a BECCS that is carbon neutral • The utilization of an oxy-fuel combustion modification to lessen costs and improve efficiency • Produce steam cycles with a high level of effectiveness • NO_x production is minimized because there is no nitrogen present • This method attains a 60–70% decrease in NO_x emissions in comparison to air-fired combustion • Has the potential to operate at high pressure, while using less energy for CO₂ compression • High temperatures complicate the combustion of pure oxygen • Requires more oxygen than pre-combustion, resulting in higher expenses for both capital costs and energy use • The air capture unit needs much energy and it significantly influences the overall efficiency of power plants • The development of alternatives to cryogenic distillation for air separation units • The expense of flue gas recirculation and air separation dramatically lowers the economic advantage • Construction materials must endure the elevated temperatures of combustion gases produced by the high oxygen concentration during fuel combustion • Technology must be tested before being used in huge operations • High potential for CO₂ leakage	• High temperatures complicate the combustion of pure oxygen • Requires more oxygen than pre-combustion that is more expensive in terms of not only capital costs but also energy use • The air capture unit needs much energy and it significantly influences the overall efficiency of power plants • The development of alternatives to cryogenic distillation for air separation units • The expense of flue gas recirculation and air separation dramatically lowers the economic advantage • Building materials must be capable of withstanding the high temperatures of the combustion gases that result from the high oxygen concentration during fuel combustion • Technology must be tested before being used in huge operations • Great potential for CO₂ leakage	• Pilot and demonstration projects • Some commercial-scale facilities	Cabral et al. (2019); Shiyi Chen et al. (2019); Elhenawy et al. (2020); Mondal et al. (2012); Olajire (2010); Omodolor et al. (2020); Wei et al. (2020); K. Zhao et al. (2023)

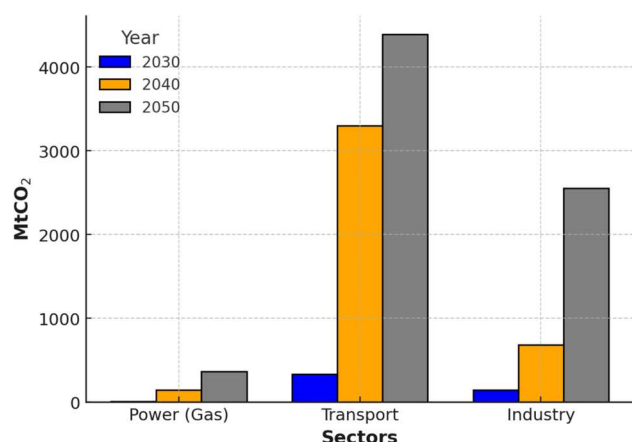


Fig. 7 Global annual CO₂ DAC capacity demand by sector (Fasihi et al. 2019)

vulgaris, *Scenedesmus obliquus*, and *Scenedesmus* sp., can metabolize carbon dioxide and create the finest lipids that are necessary for the manufacture of biodiesel (Lam et al. 2012). Open ponds are the most prominent type of MA cultivation, and they are also employed commercially. For the development of microcultures, this sort of artificial pond should have a wide surface area and a minimal depth of between 0.3 and 0.5 m. Another way to cultivate microalgae is in photobioreactors (PBRs). Due to the abundance of CO₂ and light, PBRs are one of the crucial parts of MA manufacturing. For this, both closed and open systems are utilized (Lam et al. 2012). Figure 8 is a schematic representation of carbon capture and utilization through algal bio-reactors. The process begins with CO₂ emissions from a coal-fired

power plant, combined with nutrients and water, introduced into photo-bioreactors under the influence of light. The algae produced undergoes harvesting and dewatering, followed by liquid extraction. The resulting algal oil is upgraded to biodiesel, while the residual algal biochar is applied to soil as a fertilizer, showcasing a sustainable pathway for CO₂ sequestration and resource recovery.

Carbon dioxide capture by nanotechnology

The deliberate manipulation of materials at the 1–100-nm nanoscale is recognized as nanotechnology (Yave et al. 2010). It produces a few noteworthy benefits, including distinct physiochemical characteristics and a high active surface space-to-volume ratio. Nanomaterials are divided into four distinct groups, including nanocomposites, nanoporous materials, nanocrystalline materials, and nanomaterials with hollow shapes. Compared to other sorbents, nanomembranes have a significantly greater sorption rate since they are formed using a variety of precursor nanomaterials. The targeted gas for adsorption, however, affects the membrane's permeability and selectivity. Nanomaterials feature mesoporous silica nanoparticles, carbon nano-adsorbents, metal–organic frameworks (MOF) containing embedded nanoparticles, and nanoscale zeolites. Alkanes, CO₂, and water can be captured through pores with a size of 4–10 nm. Zeolites are alkaline earth and alkali elements-containing crystalline, porous, 3-D, highly organized aluminosilicates with pores measuring between 0.5 and 1.2 nm (I. U. Khan et al. 2018a, b). Owing to their elevated adsorption capability, they are commonly employed for the purpose of extracting CO₂ from the atmosphere, resulting in a range of

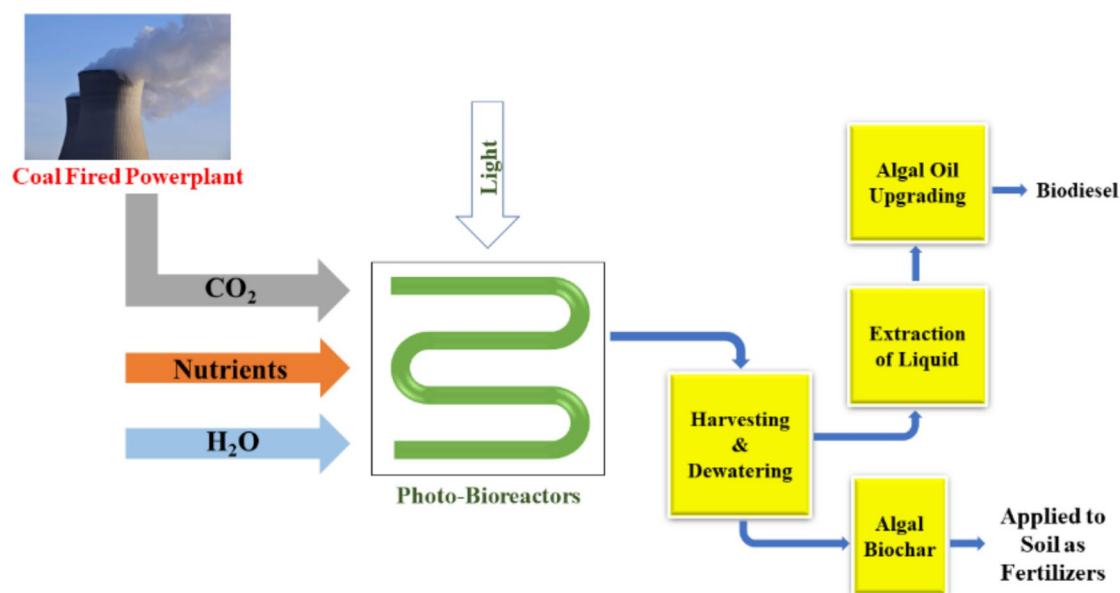


Fig. 8 Flow diagram of carbon dioxide (CO₂) capture by algae, adapted from Anwar et al. (2018)

4–300 mg of CO₂ per gram under optimal conditions (Fashi et al. 2021). Zeolites are altered on a nanoscale. Due to the modification's increased surface zone, which in turn contains higher adsorption-active regions, the sorption capacity is boosted (Chang et al. 2020). Metal cations serve as the main atom in MOF, which gets attached to other organic molecules to form a 3-D crystalline structure. To boost the structure's capacity for CO₂ adsorption, several changes are performed, such as the addition of carbon nanotubes, which are nano-hollow structures with a diameter ranging from 0.8 to 2 nm and a length of less than 100 nm and transforming the pore surfaces with amines (I. U. Khan et al. 2018a, b). Adsorption of CO₂ occurs mostly through Fe₃O₄ nanoparticles rather than multiwall nanotubes. A substantial fraction of CO₂ absorption in nanocomposites is enhanced by the chemical interaction between CO₂ and iron oxide nanoparticles (Babaei et al. 2017). Numerous studies have emphasized the restriction on temperature increases that go beyond a specific point as they degrade the adsorption capacity of nanoparticles and nanomaterials (Al Atrach et al. 2024). Since the synthesis of nanoparticles is an expensive and difficult procedure, focus should be placed on the most economically viable method (R. Kumar et al. 2020a, b).

Carbon dioxide capture by using biochar

Among the materials for CO₂ collection, biochar is a strong contender. It offers a greater selection of raw materials than other adsorbent materials and has less of a detrimental impact on the environment (Guo et al. 2022). The enormous potential for biochar application in various areas to reduce as much as 2.56* 10⁹t CO₂ of total greenhouse gas (GHG) emissions annually, which accounts for 5.0% of the world's GHG emissions (L. Wang, Deng, Yang, Hou, & Hou, 2023). Biochar will sequester CO₂ from the environment as carbonate minerals. Furthermore, the incorporation of biochar in construction of buildings and roads can reduce GHG by 25% (Gupta et al. 2018). To boost pure biochar's ability to absorb CO₂ and stop global warming, both KOH and KOH + CO₂ activation can be applied (Dissanayake et al. 2020).

Charcoal as a carbon adsorbent

Charcoal can be made almost anywhere in the world. It is used to neutralize harmful substances, clean food, transport chemical catalysts, enrich soil, reduce GHG, and trap sound, microwave, and radio waves, among other things (Emrich 2013). Activated charcoal has been created from a variety of natural biowaste to remove different dyes. In addition, wheat straw charcoal, and brick kiln charcoal, respectively, had adsorption capabilities of 80.88 and 72.81 mg/g. For Acid Red 119, 44.48 mg/g of pea shell charcoal was used, while for Acid Blue 15, 126.53 mg/g of tea leaves was used.

The highest sorption capacities for charcoal made from groundnut shell, cow dung, and bagasse were determined to be 100.57, 50.30, and 38.32 mg/g, respectively (Bello et al. 2015). An essential component in capturing carbon is charcoal, which is easily made in a small kiln. The temperature is maintained with an induction heater and is carefully monitored during pyrolysis using a thermal sensor. Typically, high temperatures are used to carry out carbonization and activation. It is subsequently grounded and let cool down at ambient temperature. This activated charcoal absorbs CO₂ by adsorption. In the absence of a vacuum, pre-absorption is carried out to assure 100% CO₂ circulation at a pressure of 40 cm Hg. The relevant sample is then subjected to gas chromatography (GC) analysis. In the presence of a vacuum, post-absorption is conducted, and GC analysis is performed. The differences between the peaks on the chromatogram provide information about the amount of CO₂ absorbed throughout the procedure (P.-H. Huang et al. 2015).

Carbon dioxide transport

It is necessary to transport the captured CO₂ to suitable storage facilities after it has been captured, which could be depleting oil and gas deposits, saltwater aquifers, or geological formations. Trucks, ships, or pipelines are typically utilized for this purpose, depending on the volume and distance of CO₂ that needs to be delivered. The most popular and economical means of long-distance transportation is through pipelines. But careful planning, oversight, and regulatory control are required to guarantee the safe and efficient transportation of CO₂ in order to minimize environmental issues and stop leaks. Furthermore, the development of CO₂ capture and transport infrastructure and laws is necessary to both scale up these technologies and achieve major global reductions in CO₂ emissions (McLaughlin et al. 2023).

Pipeline

The most economical, energy-efficient, and ecologically responsible way to move big amounts of CO₂ across long distances is by pipeline (Roussanaly et al. 2013). CO₂ transportation pipelines move captured CO₂ from its sources such as factories or power plants—to a storage location, which is usually a depleted oil and gas reservoir or an underground geological formation. CO₂ pipes can also provide CO₂ to locations for industrial processes like enhanced oil recovery (EOR). The expansion of pipelines is motivated by the necessity to decrease emissions and alleviate the impacts of climate change. The need for building CO₂ pipelines has grown dramatically and is anticipated to do so in the years to come (Paluszny et al. 2020). In order to reach carbon mitigation targets by 2050, 10 billion tons of CO₂ must be

carried across the proposed 200,000 km pipeline route (Tong et al. 2019).

CO₂ can be transported through pipelines in various states, including gaseous, liquid, dense phase, and supercritical, contingent upon the employed temperature and pressure controls. According to energy costs and economic value, solid states are best suited for long-distance transportation, dense-phase and supercritical for short-distance transit, and gaseous and liquid states for overall transportation (Paluszny et al. 2020). Temperatures between 13 and 43.8 °C and pressures between 9 and 15 MPa are typical pipeline working conditions (Adu et al. 2019).

Length, capacity, and power needs are some of the variables that affect pipeline design and are typically related. The pipeline's capacity and power requirements increase with its length. Due to its affordability, carbon steel is the predominant material employed for constructing CO₂ pipelines. In shorter pipelines, CO₂ is carried in a liquid or dense phase state; but in regard to longer pipelines, it is transported as a gas. Because the energy needed to sustain phase state may vary depending on the route and environment the pipeline will traverse, these considerations must be evaluated during the pipeline design process (Roussanaly et al. 2013). Table 6 presents a range of key parameters, including pipeline length, wall thickness, external diameter, designed capacity, compressor capacity, and pressure limits (maximum and minimum). These attributes highlight the engineering considerations for the efficient and safe transport of CO₂ in carbon capture and storage systems, as reported by Noothout et al. (2014).

Pipelines carrying CO₂ carry similar risks as pipelines carrying natural gas, oil, and water. Infrequent but serious CO₂ leaks can have an effect on nearby populations, posing hazards to the removal of emissions and the health of nearby humans and animals (Teng et al. 2021). There were 46 CO₂ pipeline failure occurrences in the US between 1972 and 2012; however, no fatalities or serious injuries were reported (Adu et al. 2019). A total of 1746 people and 3500 animals perished in 1986 as a result of a large CO₂ leak from a limnic eruption in Lake Nyos (Teng et al. 2021). Embrittlement, fracture, and corrosion can all result in pipeline leaks (Lu

et al. 2020). According to pipeline corrosion research, the rate of corrosion in a pure CO₂ stream is almost negligible. Pure CO₂ consistently does not corrode when it interacts with different pipeline materials, indicating that the primary danger is contaminants associated with the captured CO₂ (Lu et al. 2020; Sun et al. 2019). A CO₂ stream's purity is mostly dictated by its source and the method used for capture (Noothout et al. 2014). The present research focuses on impurities including CO, H₂S, H₂O, O₂, and N₂ (Teng et al. 2021). These contaminants' corrosion mechanisms are often disputed and poorly understood. Although there is a lot of research on natural gas pipeline corrosion, novel reactions and situations are created by low pressure and elevated CO₂ concentrations that need to be taken into account. Impurities can raise operating energy requirements, jeopardize pipeline integrity, and alter the critical point on the phase diagram. Impurities can be kept out of pipelines by improving cleaning technology, and their effects on pipeline material can be lessened by coating technologies (McLaughlin et al. 2023). Water can worsen corrosion in addition to pollutants by generating carbonation and CO₂-hydrates. Water contamination in supercritical CO₂ moisture resulted in the corrosion of carbon steel pipeline material, although copper and aluminum did not. Iron carbonate (FeCO₃) and carbonic acid (H₂CO₃) are the products of CO₂'s attack on iron (Fe) when it dissolves in water (Adu et al. 2019).

Because CO₂'s critical temperature is so near to room temperature, environmental factors can cause it to undergo phase changes. Shifts in height and ambient air temperature can cause temperature shifts or pressure drops in pipelines. Conditions can be tracked using the gas void fraction, which characterizes gas–liquid two-phase flow. More accurate measurements of gas–liquid ratios are now possible because of developments in laser transmittance technology (H. Wu & Duan 2019). Pipeline pressure can be maintained by adding compression stations; greater diameters necessitate fewer stations. For the most cost-effective design option, appropriate modeling for specific sites is necessary because both recompression stations and greater pipeline diameters are more costly (Adu et al. 2019).

Shipping

The location of the capture and end-use sites determines the performance of ship carriers, which offer a flexible and affordable way to carry lesser CO₂ volumes across large distances. When collection and storage sites are scarce in the early stages of carbon capture, utilization, and storage (CCUS), shipping will be essential. A 30-year example study that contrasted ship carrier and onshore pipeline CO₂ transport found that ship transportation requires less initial investment but uses more utilities and is more climate-intensive (Roussanaly et al. 2013). Ship transit is more economical

Table 6 Physical attributes of pipelines carrying CO₂ (Noothout et al. 2014)

Parameter	Range
Length (km)	1.9–808
Wall thickness (mm)	5.2–27
External diameter (mm)	152–921
Capacity designed (Mt/y)	0.006–28
Compressor capacity (MW)	0.2–68
Pressure max (bar)	21–200
Pressure min (bar)	3–151

than offshore pipes for distances greater than 350 km and onshore pipelines for distances greater than 1100 km, according to a comparative economic analysis (Decarre et al. 2010). The main consumers of small-scale liquified CO₂ transportation, which usually ranges between 800 m³ and 1000 m³, are the food and beverage sectors (Al Baroudi et al. 2021).

Liquefied natural gas (LNG) and liquefied petroleum gas (LPG) shipping processes are fairly comparable to those for CO₂. Most likely, the CO₂ was transported directly to the liquefaction plant from the collection plant (Fig. 9) (Engel & Kather 2018). The process of compressing and cooling the incoming CO₂ stream is known as liquefaction. 77% of energy needs and 54% of costs are attributed to the shipping operation, which is an extremely expensive and energy-intensive procedure (Al Baroudi et al. 2021). Low pressure and temperatures about −50 °C are usually necessary for stable liquid CO₂ transfer. Impurities can have a big effect on the liquefaction process, which raises the energy requirements. Generally speaking, liquefaction procedures fall into one of two categories: open-cycle or closed-cycle refrigeration procedures. Although CO₂ is continuously captured and liquefied by liquefaction plants, ship carrier transit requires intermediate storage tanks in between trips due to its discontinuous character. Although this improves flexibility, it also makes scheduling more difficult and expensive. Because of its great modularity, this method requires careful planning and efficient communication to scale up successfully (Al Baroudi et al. 2021; Decarre et al. 2010). Large LNG and LPG boats can hold up to 270,000 m³, but it would be expensive to adapt CO₂ tanks. Due to technical issues and shifting conditions for huge amounts of liquid CO₂, existing small-scale tanks cannot be built up (Al Baroudi et al. 2021).

Although ship carriers are economical modes of transportation, their carbon footprint should not be overlooked to preserve the integrity of CO₂ transportation. Shipping necessitates energy-intensive operations, in contrast to pipelines, hence increasing environmental costs. A correlation

exists between elevated greenhouse gas emissions and vessel operations (Al Baroudi et al. 2021). Furthermore, issues may arise during proper functioning; vessel travel carries a number of risks, including collision, fire, and stranding. These mishaps would compromise the mission's intrinsic objective, and the massive amounts of CO₂ released might harm the atmosphere, water, and ship personnel. The possible consequences of widespread liquid CO₂ leakage in shipping transportation need more investigation (Al Baroudi et al. 2021).

Railway and truck

For short distances and modest CO₂ amounts, CO₂ transportation by truck or train may be more cost-effective (Sanchez et al. 2018). Truck and rail have limitations not only in terms of scale but also in terms of accessibility and route. Since CO₂ is regarded as a hazardous material, routes are limited to places where local residents are not at risk (Al Baroudi et al. 2021). Small amounts of liquefied CO₂ are frequently transported to ship carrier port terminals using trucks as an intermediary technique. For short distances, small volumes (100 kt CO₂/a), liquefaction facilities close to truck pick-up locations, or in situations where pipes are not available, trucks are preferable over pipelines (Psarras et al. 2017). Rail is a cost-efficient option when the origin and destination are at rail terminals and the distance does not exceed 700 miles (Psarras et al. 2017). Rail transport offers lower upfront capital alternatives due to existing infrastructure, but it also faces risks like truck transportation (Psarras et al. 2017).

Despite their limited use, rail and truck transportation may play a significant role in small-scale transportation during the initial stages of the expansion of carbon capture and sequestration (CCS). Because of economies of scale, it is projected that megatons of CO₂ will be transferred annually via pipelines and ships at significantly lower costs (Table 7).

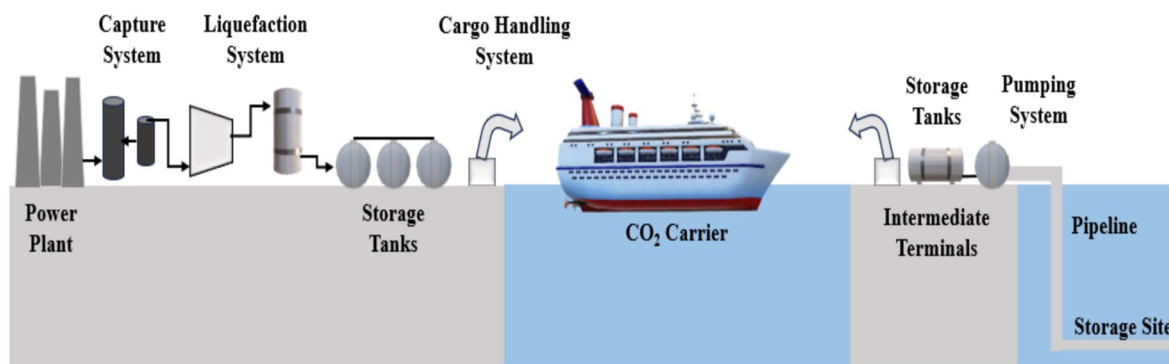


Fig. 9 The chain of the CO₂ shipping process consists of several modules (Adapted from McLaughlin et al. 2023)

Table 7 The expenses associated with CO₂ transportation, encompassing liquefaction, alongside an analysis of the merits and drawbacks of several transportation options

Transportation type	Capacity Mt CO ₂ /a	Distance km	Cost €/tCO ₂	Advantages	Disadvantages	Development stage	TRL	References
Train	1.46	598	7.3	• Lower emissions compared to shipping and roads • Large capacity • Energy efficient	• Limited geographic flexibility • Infrastructure Constraints • Schedule-sensitive and availability-tracking	• Mature	9	Dziejarski et al. (2023); McLaughlin et al. (2023); K. Zhao et al. (2023)
Truck	15–20 × 10 ^{−6}	> 100	13	• Good for short distances • Geographic flexibility • Easier to implement	• High emissions • Traffic congestion limited capacity	• Mature	9	
Shipping	2	750	11.1	• Scalable • Flexible options for routes • Cheaper for long-distance travel	• Moderate emissions • Geopolitical factors sensitivity • Port infrastructure limits	• Mature	9	
	2.5	180	13.5					
	2.5	1500	19.8					
	20	180	11.1					
	20	1500	16.1					
Offshore pipeline	2.5	180	9.3	• High efficiency • Low emissions • Capable of being integrated with CO₂ capturing facilities	• High initial investment • Limited geographic flexibility • Leakage potential and environmental dangers	• Mature	9	
	2.5	1500	51.7					
	20	180	3.4					
	20	1500	16.3					
Onshore pipeline	0.73	100	6.8					
	0.73	500	43.6					
	2.5	180	5.4					
	7.3	100	1.5					
	7.3	500	9.8					
	20	180	1.5					
	20	750	5.3					

Carbon dioxide storage processes

The hauling of CO₂ can be done in different ways, like ships, roads, and pipelines. CO₂ storage depends on the amount of CO₂ upon compression (Leung et al. 2014). CO₂ storage is very much needed to reduce emission reduction as well as global warming (Bai et al. 2024). Assuming 0.1% leakage per year, CO₂ can be put away under the soil for thousands of years. Certain factors, including pore fluid dynamics, fractures, geochemistry, and mineralogy, can affect how CO₂ behaves in the reservoir. Certain factors, including pore fluid dynamics, fractures, geochemistry, and mineralogy, can affect how CO₂ behaves in the reservoir. The storage site should meet certain requirements, such as being porous, permeable, and having rock capping that is impermeable, in order

to accommodate large-scale and long-term CO₂ storage. Stratigraphic, or structural, trapping is the most prevalent method of CO₂ trapping among all other methods. Unless there are defects, slow diffusion is controlled by injecting CO₂ into the caprock within the high-density free form. It only diffuses slowly or is absorbed into the caprock through faults. As a result, carbonates are produced that have a long lifespan (Adu et al. 2019; Amouroux et al. 2014; Johnsen et al. 2011; Z. Zhang & Huisinigh 2017). Figure 10 illustrates four key methods for CO₂ storage: subterranean geological storage, ocean storage, terrestrial storage, and mineral carbonation storage. Each approach represents a viable strategy for long-term CO₂ sequestration, contributing to the mitigation of greenhouse gas emissions and supporting climate change mitigation efforts.

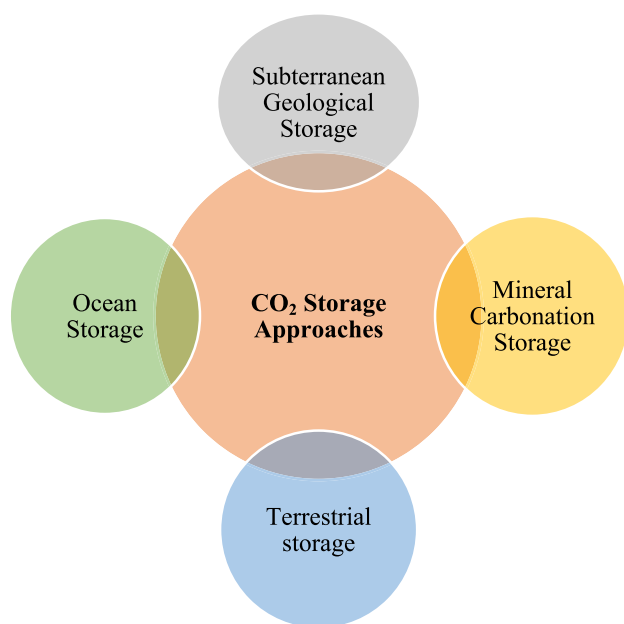


Fig. 10 Different approaches for carbon dioxide (CO₂) conveying and storage

Subterranean geological storage

Geological storage is likely better for long-term CO₂ storage due to site accessibility, cost concerns, and a host of other issues. Geological storage, or geologic carbon storage, is the act of introducing carbon dioxide into geological formations such as saline aquifers, deep underground coal seams, or exhausted oil and gas reservoirs (Ajayi, Gomes, & Bera, 2019). In order to make injection and transportation into geological formations easier, CO₂ collected from industrial sources, such as power plants, refineries, and cement factories, is compressed to a supercritical level. The compressed CO₂ is then injected into geological formations for long-lasting preservation (S. Kumar et al. 2020a, b). Carbon dioxide is forced into the formation by a meticulously drilled injection well. The efficacy and safety of the carbon storage facility are regularly assessed by monitoring the stored CO₂'s pressure, temperature, and movement (S. Kumar et al. 2020a, b). The prolonged underground sequestration of CO₂ in geological formations restricts its atmospheric emission, thus decreasing greenhouse gas emissions (Manceau et al. 2014). Large volumes of CO₂ can be retained by geological formations characterized by elevated porosity and permeability. Because of its accessibility and enormous storage capacity, geological storage is a prevalent method in storing carbon (Pan et al. 2021; Pan et al. 2022).

Present geological structures such as saline aquifers and exhausted oil and gas reservoirs are typically where carbon storage occurs. Deep saline aquifers, which are made of permeable rock, contain extremely salinized water. These

formations can retain CO₂ because they are widespread and have a lot of storage space (S. Kumar et al. 2020a, b). Following injection into the aquifer, CO₂ either becomes retained in the rock's pore spaces or dissolves in the salty water. Hydrocarbons may become trapped in depleted oil and gas sources throughout geological time spans. In well-characterized reservoirs, CO₂ injection infrastructure is often already in place, which can enhance oil or gas recovery processes and maintain reservoir pressure (Paluszny et al. 2020). Unmineable coal seams, basalt formations, and sub-seafloor sediments are increasingly being studied for carbon storage due to their unique characteristics and diversification potential. Unmineable coal seams refer to coal resources that are excessively deep, insufficiently thick, or too valuable to extract. Injecting CO₂ into these seams allows for the displacement of methane as it adsorbs onto the coal matrix (N. Zhao et al. 2018). Through a process known as enhanced coal bed methane (ECBM) recovery, this recovered methane can be used as a source of fuel. Magnesium, calcium, and iron silicate minerals are abundant in basalt rocks. Through a process known as mineral entrapment, stable carbonate minerals can be produced by the reaction between injectable CO₂ and these minerals (Raza et al. 2022). The availability of appropriate basalt rocks may restrict the implementation of this technique, despite the fact that it offers long-term, safe CO₂ storage. Significant volumes of CO₂ can be retained by sub-seafloor sediments, particularly in deep ocean basins (Luo et al. 2023).

Geological formations having a substantial storage capacity, for example saline aquifers, underground coal seams, exhausted oil, and gas reservoirs, can store significant amounts of CO₂ emissions (Aminu et al. 2017). Established industries including steel, cement, oil and gas, and electricity generation can continue to function while lowering their carbon footprints thanks to geological storage (Rissman et al., 2020). Geological storage could be implemented by developing nations to mitigate greenhouse gas emissions and stimulate economic expansion (Lee et al. 2018). However, CO₂ leakage risk in the atmosphere or other subterranean regions makes geological storage dangerous for the environment (Liu and Ramirez 2017). In geological storage, it can be costly to install and operate monitoring systems and injection wells. Geological storage could help existing enterprises and maintain fossil fuel consumption while substantially minimizing climate gas emissions. Geological storage is expected to spread into novel sectors exceeding conventional oil and gas production and energy generation (K. Zhao et al. 2023).

Mineral carbonation storage

Mineral carbonation storage, referred to as carbon mineralization, is the process of sequestering carbon dioxide and transforming it into stable, solid carbonate minerals,

such as calcium or magnesium carbonate (Snæbjörnsdóttir et al. 2020). This method offers long-lasting, secure CO₂ sequestration with a lower possibility of leakage because carbonates are environmentally benign and chemically stable. Mineral carbonation storage may occur gradually as a natural process. To effectively tackle climate change, the process must be hastened. In situ mineralization is one technique for preserving mineral carbonation; this entails the direct injection of CO₂ into certain rocks, where it interacts with minerals to form carbonates. Peridotite and basalt are the most common rock types for in situ mineralization due to their high concentration of calcium and magnesium silicate minerals (Snæbjörnsdóttir et al. 2018). Ex situ mineralization is the process of mining, crushing, and reacting to minerals with CO₂ in controlled environments, such as reactors or storage facilities (Yadav, Mehra, & study, 2021). Even though it allows for more control over reaction conditions, the processing, mining, and transportation of minerals can be energy intensive. Cement kiln dust, steel slag, and fly ash are examples of industrial waste products that are high in calcium or magnesium and can be utilized for CO₂ mineralization (Zhang et al. 2022). Utilizing waste materials provides sustainable waste management by reducing the environmental impact of extracting and processing fresh resources. Electrochemical mineralization is the process of accelerating mineral carbonation processes using electrochemical methods (Zhang et al. 2020). The selective synthesis of CO₂ into stable carbonate crystals can be accomplished with the application of an electric current. Various bacteria can enhance mineral carbonation processes by precipitating carbonate minerals or modifying the surrounding environment to encourage mineralization (Munda et al. 2018). Supercritical CO₂ mineralization refers to the transformation of CO₂ into its supercritical state, characterized by distinct features that can improve mineral carbonation processes (Q. Yuan et al. 2022). Supercritical CO₂ increases the efficiency of mineral dissolution by accelerating the process and raising overall efficiency.

By combining with carbon capture technology and turning CO₂ into stable carbonate minerals, mineral carbonation storage can lower CO₂ emissions in a number of industries, including the production of cement (Woodall et al. 2019). Concrete qualities can be improved by CO₂, which can also be permanently stored. Using waste products like slag, mineral carbonation transforms steel mill–released carbon dioxide into stable carbonate minerals (Liu et al. 2021). Fly ash and cement kiln dust are two examples of industrial trash that can be employed for CO₂ mineralization, providing a safe, long-term, chemically stable, and eco-friendly option (Snæbjörnsdóttir et al. 2020). Particularly in their early phases, mineral carbonation storage methods can be expensive, and their widespread adoption may require additional incentives or regulations (Kazemifar 2022). Growth and innovation in

mineral carbonation storage are anticipated to be fueled by research and development, with an emphasis on improving process conditions, lowering energy usage, and speeding up reaction rates. More effective and economical solutions could result from developments in supercritical CO₂ mineralization, microbial mineralization, and electrochemical mineralization. The integration of efficient carbon capture technology and mineral carbonation storage is expected to significantly reduce CO₂ emissions across several industries, including steel, cement, and power generation (K. Zhao et al. 2023).

Terrestrial storage

The process by which CO₂ emissions are taken up and stored in terrestrial ecosystems, such as wetlands, grasslands, and forests, is referred to as terrestrial storage or terrestrial carbon sequestration. The basis of terrestrial storage is the development and maintenance of biomass, or the carbon stored in biotic entities and botanical substances (Cheah et al. 2016). One step in the CO₂ collection process is the development and preservation of plants. Examples of management of land usage strategies that can improve biomass and carbon storage in terrestrial ecosystems include reforestation, afforestation, and sustainable forest management (Brack & King 2021). To ensure its effectiveness and safety, the carbon storage site's land-use practices, carbon stocks, and biomass are regularly monitored to prevent carbon from reentering the atmosphere (Tushar et al. 2024). Terrestrial storage reduces the amount released into the atmosphere and, as a result, greenhouse gas emissions by storing CO₂ emissions in soil and live biomass (Nunes et al. 2019).

Terrestrial storage practices such as afforestation, reforestation, agroforestry, conservation agriculture, wetland restoration, and grassland management reduce atmospheric CO₂ levels and slow down climate change by enhancing ecosystems' ability to store carbon. Common strategies to improve carbon sequestration and storage in terrestrial ecosystems include afforestation, or planting new trees, and reforestation, or reforesting previously wooded regions (Nave et al. 2019). Incorporating trees into agricultural areas improves crop yields, decreases soil erosion, increases biodiversity, and increases carbon sequestration (Chapman et al. 2020). Crop rotation, preserving soil cover, and reducing soil disturbance are agricultural techniques that can improve soil carbon absorption and lower CO₂ emissions from agricultural operations (Northrup et al. 2021). One sustainable grazing method that can increase grassland productivity and soil carbon sequestration is rotational grazing (Abdalla et al. 2018). Biochar, a carbon-rich byproduct of biomass pyrolysis, can enhance soil fertility and boost carbon storage in agricultural fields, while being used less frequently for terrestrial storage (Amoakwah et al. 2020). Seagrass meadows, salt marshes, and mangroves are examples of

coastal ecosystems that can be preserved and regenerated to significantly contribute in carbon sequestration and storage (Wedding et al., 2021). Urban green spaces can enhance air quality, lower CO₂ emissions, and mitigate the urban heat island effect (Pregitzer et al. 2022).

Although terrestrial storage is an affordable option for storing carbon, it could not be sufficiently scalable to lower greenhouse gas emissions over time on a global scale (Nolan et al. 2021). Carbon storage in terrestrial ecosystems can be difficult to monitor and validate since it is hard to quantify and ensure its persistence across time (Morecroft et al. 2019). Land-use change, natural disasters, and competition with other land uses can all reduce the effectiveness of terrestrial storage as a long-term carbon storage alternative (Memmah et al. 2015). Research on the production of genetically modified crops, soil amendments, and biochar can increase the potential for sequestering carbon to fight climate change by improving the efficiency of terrestrial storage. Examples of sustainable land management practices that can boost agricultural regions' ability to trap carbon while simultaneously boosting resilience and productivity include agroforestry, conservation agriculture, and rotational grazing (K. Zhao et al. 2023).

Ocean storage

Ocean storage, also known as ocean carbon sequestration, is the process by which CO₂ emissions are absorbed and stored in the water (Connelly et al. 2022). This practice is used for storing CO₂ immediately in the ocean as a solid hydrate, liquid, or dissolved gas and then transporting it to specific position in the ocean via pipelines or ships in order to reduce greenhouse gas emissions (Rau and Baird 2018). Ocean storage offers a potential solution for carbon storage due to its vast capacity and capacity to hold CO₂ for a few hundred or many years (Aminu et al. 2017). Ocean storage offers several advantages because it does not require land or resources and can be done naturally through direct injection or ocean fertilization. Ocean storage technologies raise biological concerns such as the possibility of marine ecosystem damage due to ocean acidification, notwithstanding their potential to store carbon. Ocean storage is being investigated as a possible strategy to mitigate climate change, but widespread adoption is being hampered by concerns about its long-term effectiveness and potential environmental harm. Direct injection is a common ocean storage method that releases CO₂ into the water at depths over 1000 m after being extracted from emission sources like power plants (Allen et al. 2015). Along with state-of-the-art technologies, new methods like as marine permaculture, artificial upwelling, and fertilization are being researched to decrease environmental consequences and boost ocean storage capacity for carbon sequestration. Ocean fertilization brings nutrients, such as iron, to the ocean's surface, which promotes phytoplankton growth (Babakhani et al. 2022). These tiny plants absorb CO₂ through

photosynthesis, and some of their biomass sinks to the deep ocean after they die, storing carbon for a very long time. Artificial upwelling is a process that raises the uptake of CO₂ from the atmosphere and promotes the growth of phytoplankton by pumping cold, nutrient-rich water from the deep ocean to the surface (Fan et al. 2020). In order to improve pH and CO₂ absorption, alkaline materials like olivine or limestone are added to the ocean to raise its alkalinity. This could increase the ocean's potential to store carbon for thousands of years. Marine permaculture is the cultivation of plants that absorb CO₂ and store it as biomass, which can be harvested for the production of bioenergy or stored in the deep ocean for long-term carbon storage (Hurd et al. 2022).

Ocean storage increases dissolved CO₂ availability, which boosts ocean productivity. However, it also increases the danger of environmental damage from CO₂ leaks into the ocean and incurs expenses for costly injection and monitoring systems. Ocean storage's long-term safety and environmental effects are still unknown, and public opinion is problematic. Research, environmental hazards, and attention to alternative carbon sequestration techniques will determine the future picture. More sustainable and efficient solutions could result from improved ocean storage technologies (Zhao et al. 2023) (Table 8).

Monitoring carbon dioxide leakages

Many serious problems, including the acidification of the soil and ocean, are brought on by CO₂ leakage during storage or transit. Both permeable paths and cap rocks have the potential to cause leakage. The faults and fractures in the crust of the planet can cause it. In contrast, leaking through cap rock is steady and takes an extended period. It is important to keep up the cap rock's pressure both before and after injection. The region should be observed during the pre- and post-CO₂ injection processes to ensure secure and long-term storage (Leung et al. 2014). To quickly detect and assess CO₂ leakage from storage locations, verification and monitoring must be done continuously. For CO₂ storage to remain long-term and intact, several monitoring technologies are used, including the following:

- a. **Surface Monitoring:** Methods including remote sensing, satellite tracking, and ground-based surveys can find high CO₂ levels in the air close to storage facilities.
- b. **Subsurface Monitoring:** Tracking the transport of CO₂ underground and locating potential leakage channels requires the use of techniques like well logging, pressure monitoring, and seismic imaging.
- c. **Tracer Technology:** Inert tracers can be injected alongside CO₂ to help detect probable leak channels and assess the effectiveness of CO₂ storage (Friedmann 2008).

Table 8 CO₂ storage options with their advantages and disadvantages

CO ₂ storage option	Advantages	Disadvantages	Development stage	References
Geological storage	• High capacity for storing and tested technology • The possibility of enduring storage	• High initial expenses • Possibility of leakage • Appropriate geological formations are necessary	• Mature with ongoing commercial-scale projects	Zhao et al. (2023)
Mineral carbonation storage	• Possibility of long-term storage • Potential for carbon neutrality when combined with renewable energy sources	• Limited scale • The reaction rates are slow • High energy requirements • Cost and scalability challenges	• In the initial phase Including a few pilot projects	
Terrestrial storage	• Improving ecosystem services • Promotes biodiversity	• Insufficient storage capacity • Requires ongoing management • Competition for land usage	• Mature with extensive research and adoption	
Ocean storage	• Can be relatively inexpensive • Large storage capacity	• Ocean acidification concerns • Marine life's exposure to the environment • Regulatory obstacles	• Initial stage with limited implementations	

Utilization of captured carbon dioxide

Once extracted, CO₂ can be derived in many ways, such as compressing, storing, or applying. Utilization might result in financial gains that would more than make up for the price of CCSs. The most widespread uses are growing microalgae, producing chemicals, and using it as a green solvent (Godin et al. 2021). Figure 11 illustrates various utilization pathways for CO₂, including its use as a fuel feedstock, in

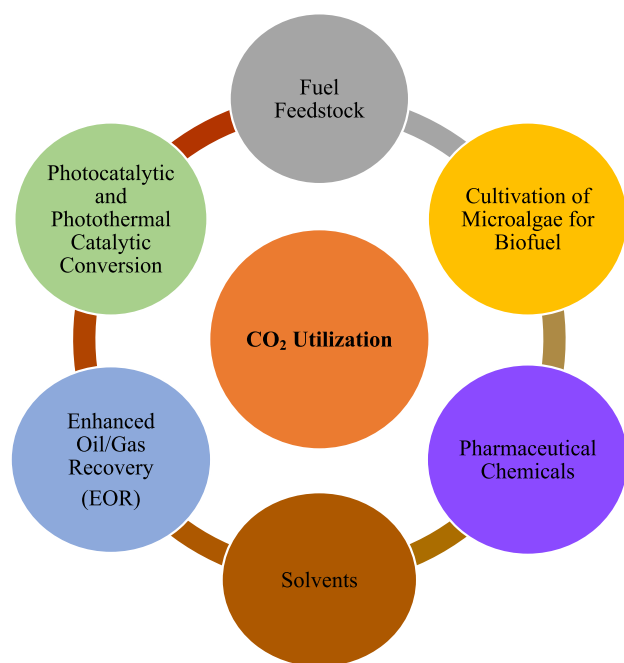
the cultivation of microalgae for biofuel production, in the synthesis of pharmaceutical chemicals, as solvents, in EOR, and through photocatalytic and photothermal catalytic conversion processes. These applications demonstrate the potential of CO₂ utilization to contribute to sustainable practices and resource recovery (Alper and Orhan 2017; Valluri et al. 2022).

Carbon dioxide uses as fuel feedstock

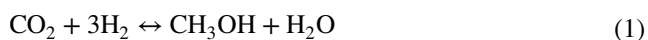
Because of their economic worth and prospective applications in a range of industries, including fuel cells, transportation, and power generation, carbon dioxide-derived fuels are regarded as the most attractive option for decarbonization. Energy dependence on fossil fuels would be lessened by turning CO₂ into fuels, which would also cover the cost of capture and promote commercialization. Utilizing CO₂ as energy carriers for transportation, dimethyl ether, Fischer–Tropsch, formic acid, methane, and methanol fuels have all demonstrated possibility (Al-Mamoori et al. 2017; Hepburn et al. 2019; Thonemann & Pizzol 2019).

Methanol

Due to its exceptional combustion properties, including its high energy value of 726.2 kJ/mol, transformation to methanol enables the possibility of large-scale CO₂ utilization. This attribute renders it well-suited for utilization as a fuel for transportation purpose (Osman et al. 2021). Syngas, which is predominantly sourced from large amounts of methanol, is produced from natural gas. Through the catalytic hydrogenation of CO₂ (Reaction (1)), methanol can be generated immediately (E. Alper


Fig. 11 Different applications for captured carbon dioxide (CO₂) (Adapted from Alper and Orhan 2017; Valluri et al. 2022)

& O. Y. Orhan, 2017; Hepburn et al. 2019; Thonemann & Pizzol 2019).



Formic acid

For low-temperature fuel cells, formic acid is one possible fuel source. The two main processes for generating formic acid from CO_2 are electrochemical reduction of CO_2 and hydrogenation of CO_2 (E. Alper & O. Y. Orhan, 2017; Hepburn et al. 2019; Thonemann & Pizzol 2019).



Latest developments in catalytic devices that break down formic acid to create H_2 under more lenient conditions than those for methanol may reduce the energy needed to convert CO_2 into formic acid. CO -free H_2 is created for fuel cells using homogeneous catalysts, which also encourage the dehydrogenation process and steer clear of the decarbonylation pathway (Michele Aresta, Dibenedetto, & Angelini, 2014).

Carbon dioxide methanation

CO_2 methanation is a method that turns CO_2 into methane (CH_4), a natural gas replacement that burns cleanly. With the help of this technique, renewable hydrogen may be stored as methane, which makes it a flexible and portable energy source (Sayma et al. 2014).

Dimethyl ether

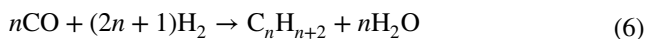
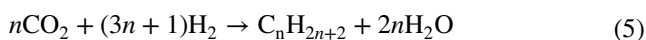
Dimethyl ether (DME) is a very efficient and environmentally friendly replacement for fossil fuels in diesel engines. Synthesis gas, methanol, CO_2 , H_2 , and other gases can all be used as precursors to create DME (Hepburn et al. 2019; Rahman et al. 2017; Thonemann & Pizzol 2019).



Fischer–Tropsch products

The Fischer–Tropsch process can be used to create liquid hydrocarbons from carbon dioxide and hydrogen. The ambient temperature of the reaction affects the makeup of the final product. For example, a naphtha byproduct and kerosene jet fuel may be produced at a higher temperature and

under specific process conditions (Choi et al. 2017; Hepburn et al. 2019; Thonemann & Pizzol 2019).



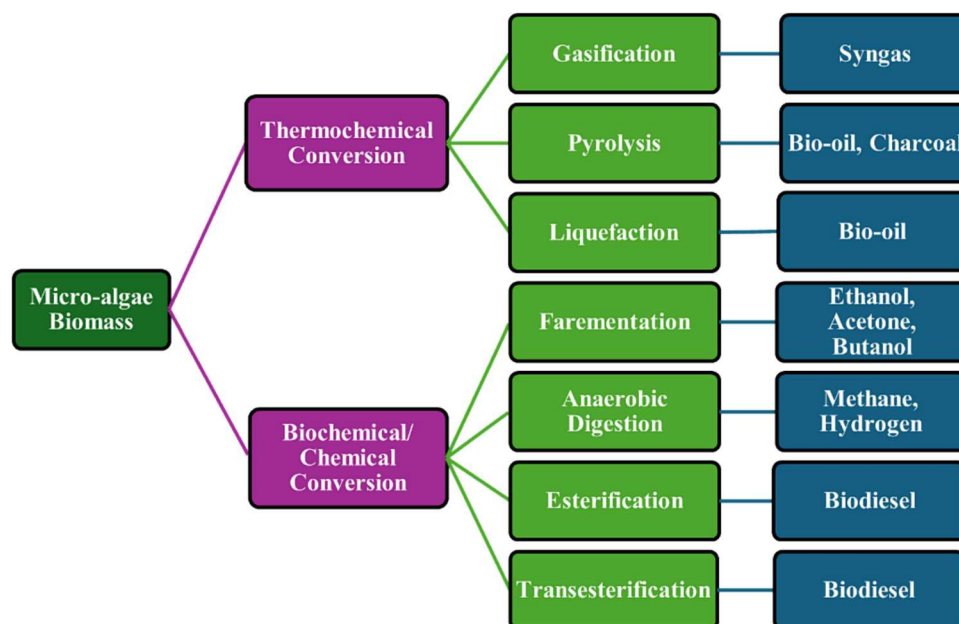
A major obstacle to the broad usage of carbon dioxide as a combustion fuel is its thermodynamic stability. Because of the high temperatures and energies required for CO_2 conversion, catalysts are essential for achieving high fuel yields (Aresta et al. 2014). The creation of novel catalysts will facilitate fuel production using CO_2 . If the price of fossil fuels increases globally, alternative fuel production methods—like encouraging renewable energy sources including geothermal, wind, and solar—will become more alluring. This makes it possible to use green energy, which emits no carbon dioxide, to convert CO_2 into extremely stimulating fuel sources (Wang et al. 2017). One of the primary byproducts of the Fischer–Tropsch synthesis of collected CO_2 is synthetic fuels (SFs). In many applications, these fuels—often referred to as “carbon-neutral” or “low-carbon” fuels—potentially replace petroleum-based fuels. The use of renewable feedstock (CO_2) in the production of SFs from CO_2 capture has the benefit of reducing dependency on fossil fuels and decarbonizing transportation. It has been demonstrated that the Fischer–Tropsch process, which employs atmospheric CO_2 , can create synthetic hydrocarbons like jet fuel, diesel, and gasoline (Smith 2019).

Cultivation of microalgae for biofuels

Microalgae are a plentiful biomass energy source that can be grown using CO_2 . The grown microalgae can be utilized by different conversion technologies to create biofuels and highly valued carbohydrates, proteins, and polymers can be grown using CO_2 (Fig. 12) (Khan et al. 2018a, b). When compared to pure CO_2 intake, microalgae may directly resolve CO_2 from waste gas streams (WGS) like flue gas, resulting in a 30% increase in biomass production (Douskova et al. 2009). Flue gas contains additional nutrients like sulfur and nitrate that contribute to this growth (Aresta et al. 2014; Lamm 2019). By producing 100 tons of algal biomass, which has a dry weight carbon content of roughly 50%, it is possible to fix about 183 tons of CO_2 (Razzak et al. 2013).

Both closed photobioreactors (PBRs) and open raceway ponds can be used for microalgae cultivation. However, because of the extensive acreage requirements and environmental instability, placing ponds close to CO_2 sources might be challenging (Lamm 2019). The productivity of

Fig. 12 Different conversion processes of microalgae biomass for biofuel production (Khan et al. 2018a, b)



algal ponds is limited by the challenges of controlling the process to adjust to variations in ambient temperatures and light utilization. Despite being easier to maintain, PBRs are more costly than algal ponds, even though they are often considered to be the most promising and successful technique for producing ethanol (Razzak, Hossain, Lucky, Bassi, De Lasa, et al., 2013).

Because of their diverse protein, lipid, and carbohydrate content, microalgae are a strong competitor for producing carbon-neutral fuels like biodiesel, bio-methane, bio-hydrogen, and bioethanol (Mathimani et al. 2019). Thermochemical conversion methods such as pyrolysis, gasification, and liquefaction are used to create solid fuels like “bio-char,” liquid fuels like “bio-oil,” and gaseous bio-fuel (Guruviah et al. 2019). Fermentation mostly yields alcohols, such as bioethanol and bio-butanol, as well as hydrogen and other compounds. On the other hand, anaerobic digestion (AD) predominantly generates methane and hydrogen. The primary chemical procedure for converting algal oil or lipid into biodiesel is transesterification. Of course, after drying, microalgal biomass can be burned to provide heat and power (Khan et al. 2018a, b). In addition to producing biofuels, microalgae may produce a wide variety of other useful items. Microalgae-derived biomass (MDB), for example, can be utilized as a raw material to create sustainable goods. Colors, proteins, and carbohydrates that are utilized in the culinary, cosmetic, and pharmaceutical industries can be extracted from biomass by processing. Because MDB is so versatile, it can be utilized to produce a diverse array of sustainable products (Raman 2016).

Utilizing carbon dioxide as a fine chemical or medicinal

Recently, carbon dioxide (CO₂) has attracted a lot of interest as a feedstock and as a sustainable method for producing specialty and pharmaceutical compounds. Through a variety of inventive procedures, CO₂, a widely accessible and plentiful carbon source, can be converted into useful chemical building blocks. This method not only helps to lower CO₂ emissions, but it also offers a viable route toward the creation of greener, more sustainable chemical synthesis.

One method that can transform CO₂ emissions into beneficial chemical compounds is carbon dioxide capture and utilization, or CCU (M. Aresta & Dibenedetto 2007). Additionally, captured CO₂ can be used to create organic carbonates, which are used in the production of polymers, coatings, lubricants, fertilizers, insecticides, and medications. While carboxylation reactions employ CO₂ to make organic molecules, reduction mechanisms break C-O bonds to make chemicals. These are two distinct conversion methods. Both reduction and carboxylation reactions break C-O bonds to create compounds, using CO₂ as a precursor for organic molecules. Substances can be changed using these reactions (Godin et al. 2021). Using CO₂ as a starting source, cyclical carbonates—important intermediates in the synthesis of polymers, solvents, and specialized chemicals—can be produced. By reacting CO₂ with epoxides, scientists have developed catalytic devices that effectively produce cyclic carbonates. Scientists have created catalytic devices that efficiently react CO₂ with epoxides to produce cyclic carbonates. When the right catalysts are used, the reaction between

CO₂ and epoxides can produce cyclic carbonates and the result is a sustainable and environmentally benign process (Liu et al. 2015).

Carbon dioxide usage as a solvent

Researchers are always exploring novel approaches to improve CO₂-based solvents, enhance their effectiveness, and broaden the industries across which they are used. The overarching objectives of green chemistry are furthered by using CO₂ as a solvent in sustainable and environmentally responsible operations. Due to its special qualities and advantages for the environment, much emphasis has been paid to carbon dioxide (CO₂) for use as a solvent in numerous applications. Under some circumstances, such as high pressure and temperature, CO₂ can behave as a solvent, enabling its use as a substitute for traditional organic solvents. This strategy aims for more sustainable and ecologically friendly processes, which is consistent with the concepts of green chemistry (Jessop 2011; Martyn Poliakoff et al. 2002). CO₂'s use as a solvent is notable for its use in the extraction and separation of natural chemicals. Food, medications, and botanicals have all been employed as sources for important compounds that have been extracted via supercritical carbon dioxide (SC-CO₂) extraction. SC-CO₂ has advantages over conventional extraction methods, including the ability to produce solvent-free extracts, high selectivity, and moderate extraction conditions (Reverchon & De Marco 2006). Because so little supercritical CO₂ (SC-CO₂) is required, utilizing CO₂ as a solvent for chemical operations is ecologically beneficial. This amount is comparable to the amount of reactant materials, unlike conventional solvents which typically require 3 to 20 times more solvent than reactant materials (Marriott et al. 2015). By replacing organic solvents with SC-CO₂, which is generally burned to dispose of waste organic solvents, additional CO₂ emissions from these procedures would be mitigated (Aresta et al. 2014). The amount of pressure needed to compress CO₂ to a supercritical condition is the primary drawback of employing supercritical CO₂ as a solvent (above 72.8 atm). When validating the energy use of SC-CO₂ with traditional solvents, it is important to take into account the pressure associated with operational costs (Godin et al. 2021).

Utilizing carbon in improved oil and gas recovery (EOR/EGR)

Carbon dioxide is delivered into hydrocarbon reservoirs through CO₂-EOR and CO₂-EGR systems to augment the extraction of natural gas and oil. By improving the extraction and utilization of oil and gas resources and storing CO₂ underground, the process reduces greenhouse gas emissions (Al-Shargabi et al. 2022; Hamza et al. 2021). In CO₂-EOR

method, carbon dioxide is injected into oil reservoirs to boost crude oil extraction, which decreases oil viscosity and improves its mobility (Y. Liu & Rui 2022). Through the injection of CO₂ into natural gas reservoirs, the CO₂-EGR technique raises reservoir pressure and enhances gas production and recovery (Omari et al. 2022).

Immiscible and miscible CO₂-EOR techniques are examples of common CO₂-EOR technologies. By adding CO₂ to the reservoir at or exceeding the minimal miscibility pressure (MMP), the miscible CO₂-EOR technique reduces the viscosity and interfacial tension of the oil (Czarnota et al. 2017). As a result, oil recovery and transportation are enhanced. Miscible CO₂-EOR frequently works well in reservoirs with light oil and good permeability. If CO₂ is introduced below the MMP or reservoir conditions preclude miscible displacement, it swells oil instead of mixing with it, increasing its mobility and decreasing its viscosity (Li-ping et al. 2015). This method is typically less successful than miscible CO₂-EOR, even though it can be used in a larger variety of reservoirs. Innovative approaches include carbonated water injections, foam-assisted CO₂-EOR, WAG (water-alternating-gas) injections, and hybrid EOR technologies which are being explored to boost sustainability and efficiency. Water and CO₂ are pumped into the reservoir in turn during the WAG injection process. Water infusion enhances oil sweep efficiency, while Carbon dioxide reduces oil viscosity and enhances mobility. WAG can be used for processes that do not mix and processes that do mix. The technique known as “foam-assisted CO₂-EOR” creates CO₂ foam by injecting CO₂ along with a foaming agent, usually a surfactant (Xu et al. 2017). Because this foam is more viscous than CO₂ alone, it decreases gas channeling and increases sweep efficiency. CO₂ is dissolved in water and then injected into the reservoir to create carbonated water (Nowrouzi, Manshad, & Mohammadi, 2020). Oil displacement is enhanced, and oil viscosity is decreased by the carbonated water. By mixing CO₂-EOR with other methods like steam, chemicals, or low-salinity water, hybrid EOR procedures increase oil recovery and can be customized for particular reservoir conditions (Mogensen and Masalmeh 2020).

CO₂-EGR is underdeveloped than CO₂-EOR, with many systems still in the experimental or pilot research stage. More funds have been allocated to oil recovery technologies like CO₂-EOR than to gas recovery systems like CO₂-EGR since, historically, the demand for oil has outstripped that for natural gas (Caineng et al. 2023). Compared to CO₂-EOR technologies, CO₂-EGR technologies have fewer field applications, which means that there is less practical experience to assist them be improved. The behavior and processes of CO₂ in gas reservoirs are generally more complex than those in oil reservoirs (Burrows et al. 2020). The intricacy of CO₂-EGR techniques and the changing composition of gas reservoirs make their development increasingly challenging. Continuous CO₂ injection is a relatively simple and commonly employed technique in CO₂-EGR technologies

that entail continuously injecting CO₂ into the gas reservoir (Bongole et al. 2019). The CO₂ injection increases reservoir pressure and facilitates the flow of natural gas into production wells.

Natural gas and oil are two of the many industries that use techniques like CO₂-EOR and CO₂-EGR. CO₂-EOR enhances recovery from declining oilfields, improving production and extending reservoir life, whereas CO₂-EGR enhances natural gas recovery from mature gas fields with a high CO₂ component (Y. Liu & Rui 2022; Omari et al. 2022). CO₂-EOR and CO₂-EGR are techniques that employ CO₂ captured from power stations or industrial operations to store CO₂ underground in order to support the CCTUS industry and reduce greenhouse gas emissions (Mikulčić et al. 2019). The drawbacks of CO₂-EOR and CO₂-EGR are that they increase production and prolong the life of oil and gas fields that may promote fossil fuel usage and contradict the shift to more sustainable energy sources (Kalair et al. 2021). As the globe shifts to cleaner energy sources and emission reduction, the growing use of CO₂-EOR and CO₂-EGR technologies by oil and gas companies speaks well for these technologies' future. Technological advancements in CO₂-EOR and CO₂-EGR will enhance monitoring systems, reduce costs, and boost effectiveness, making them more attractive and available for a variety of applications (Zhao et al. 2023).

Photocatalytic and photothermal catalytic conversion of CO₂

Novel techniques including photocatalytic and photothermal catalytic conversion of CO₂ are intended to use solar energy to drive chemical reactions that convert CO₂ into products with value additions. The US DOE's Sandia National Laboratories (SNL) has presented a working prototype of the "Sunshine to Petrol" (S2P) reactor, which is a noteworthy development (Zhu 2019). Widespread R&D is currently being conducted worldwide. The creation of effective, stable, and selective photocatalysts for the chemical conversion of CO₂ is the main area of research. To improve the stability and efficiency of photoactive materials, a number of techniques have been employed thus far, including co-catalysts, passivation layers, doping, combining two or more semiconductors, and the fabrication of nanostructured materials. Using semiconductor materials to capture solar energy and catalytically convert CO₂ into value-added chemicals like methanol (CH₃OH), carbon monoxide (CO), or methane (CH₄) is known as photocatalytic conversion of CO₂ (L. Yuan & Xu 2015). The process of photothermal catalytic CO₂ conversion uses catalytic materials that effectively transform solar energy into heat, which fuels CO₂ reduction reactions (Alper and Orhan 2017) (Table 9).

Table 9 Technical readiness level of different CO₂ products

TRL	Products	References
3	•Dimethyl ether	Dziejarski et al. (2023)
5	•Fischer–Tropsch products	
6	•Ethanol •Formic acid •Syngas	
7	•Calcium carbonate •Magnesium carbonate •CO ₂ -based polymer	
8	•Sodium bicarbonate •Methane •Dimethyl carbonate •Dry algae powder	
9	•Methanol •CO ₂ -based polycarbonates •Urea	

Discussion and future directions

CCTUS technology is not only for removing excess amounts of CO₂ from the environment but also for providing a low-pollution energy source to the environment while reusing fossil fuels from industrial emissions. The key findings of this paper have been discussed as follows:

- Utilizing carbon capture techniques including oxy-fuel combustion, post-combustion, and pre-combustion, CO₂ is removed from sources or the atmosphere. Although it requires infrastructure adjustment, pre-combustion delivers excellent efficiency. Although energy-intensive, post-combustion is flexible. High purity is captured by oxygen-fuel combustion, but energy input is necessary. Research, cost cutting, and industrial integration are essential for future success.
- There are benefits and drawbacks to CO₂ transportation via pipeline, shipping, road, and rail. While shipping gives flexibility but has modest emissions and depends on port infrastructure, pipeline transportation is reusable and feasible. Although it has a limited capacity and heavy emissions, road transportation is appropriate for short distances. Despite its lower emissions, rail transportation is constrained by its infrastructure.
- The objective of carbon storage techniques, including geological, mineral, terrestrial, and ocean carbonation, is to safely store captured CO₂. However, they have problems with cost, energy intensity, and public acceptance. Future developments are contingent upon public acceptability, regulatory frameworks, cost reduction, and research. For climate change solutions to be effective, cross-sector innovation and cooperation are essential.

- Technologies for carbon utilization are being developed for a number of uses, such as oil/gas recovery, solvents, medicinal compounds, microalgae farming, and fuel feedstock. These methods rely on local conditions and demand a large amount of energy. For broad acceptance, more study, effectiveness, and cross-industry integration are essential.

Conclusion

This review provides a comprehensive analysis of CCTUS technologies, emphasizing their critical role in achieving net-zero CO₂ emissions. By examining the current state of capture, transport, utilization, and storage techniques, this study highlights their technical advantages, operational challenges, and potential for integration into global industrial systems. While significant progress has been made, this review underscores that widespread adoption of CCTUS systems requires overcoming several persistent challenges, including economic feasibility, scalability, and technological innovation.

Key findings from this review indicate that while technologies like pre-combustion, post-combustion, and oxy-fuel combustion CO₂ capture offer varying degrees of efficiency and applicability, they remain energy-intensive and costly. Emerging methods, such as direct air capture, algae-based systems, and nanotechnology-enhanced processes, show promise in reducing energy consumption and operational costs. However, these innovations require further research to enhance their scalability and integration into existing industrial frameworks. Similarly, storage solutions, such as geological sequestration and mineral carbonation, demonstrate potential for long-term stability but require site-specific assessments and advancements in monitoring technologies to ensure safety and environmental compliance.

The transport of CO₂ remains a logistical and financial challenge, particularly for large-scale operations. Developing cost-effective pipeline infrastructure and exploring alternative transport methods, such as shipping, are essential for enhancing the feasibility of CCTUS systems. Moreover, CO₂ utilization provides an opportunity to offset costs and create value-added products, such as chemicals, fuels, and construction materials, but commercial-scale applications remain limited by market and technological constraints.

This review identifies several knowledge gaps and research priorities to advance the field of CCTUS:

1. Future research should focus on optimizing capture and storage systems to enhance scalability and reduce costs, particularly through the integration of novel materials and energy-efficient processes.
2. Economic models and incentives, such as carbon pricing, subsidies, and market mechanisms for CO₂-derived products, must be developed to encourage investment and adoption of CCTUS technologies.
3. Effective policy frameworks, supported by collaboration between researchers, environmental advocates, and policymakers, are essential to drive innovation and implementation.
4. Robust monitoring technologies and strategies are required to ensure the long-term safety and effectiveness of CO₂ storage solutions.
5. Exploring the synergy between CCTUS systems and renewable energy sources can enhance their overall sustainability and reduce dependency on fossil fuels.

In summary, CCTUS technologies offer a multifaceted solution to mitigating CO₂ emissions, but their success depends on addressing key environmental, technological, economic, and policy challenges. By synthesizing insights from existing research, this review provides a roadmap for advancing the field and fostering innovation. The findings aim to inspire a collaborative approach among researchers, industry stakeholders, and policymakers, ensuring that CCTUS systems contribute effectively to global climate change mitigation and the transition to a sustainable, net-zero future.

Nomenclatures ACs: Activated carbons; AD: Anaerobic digestion; APCC: Aqueous post-combustion CO₂ capture; Bio-CCS: Biological carbon capture and storage; CC: Carbon capture; CCS: Carbon capture and sequestration; CCU: Carbon capture unit; CCUS: Carbon capture utilization and storage; CCTUS: Carbon capture, transportation, utilization, and storage; CO: Carbon monoxide; CO₂: Carbon dioxide; DAC: Direct air capture; EGR: Exhaust gas recirculation; EGS: Enhanced geothermal system; EOR: Enhanced oil recovery; EGR: Enhanced gas recovery; GC: Gas chromatography; GHG: Greenhouse gas; IEA: The International Energy Agency's; LNG: Liquefied natural gas; LPG: Liquefied petroleum gas; LSBM: Liquid solvent-based method; MA: Microalgae; MDB: Microalgae-derived biomass; MCS: Membrane contactor setup; MIPs: Molecularly imprinted polymers; MOF: Metal-organic frameworks; NETL: National Energy Technology Laboratory; NOx: Nitrogen oxides; OC: Operational cost; OFC: Oxy-fuel combustion; OSM: Oxygen separator membrane; PBRs: Photobioreactors; PP: Power plants; Post-CC: Post-combustion capture; Post-CCC: Post-combustion carbon capture; Pre-CC: Pre-combustion capture; Pre-CCC: Pre-combustion carbon capture; RFG: Recycled flue gas; SC-CO₂: Supercritical CO₂; SFs: Synthetic fuels; SGT: Synthesis gas treatment; SNL: Sandia National Laboratories; SO₂: Sulfur dioxide; SOx: Sulfur oxides; SSBM: Solid sorbent-based method; Syngas: Synthesis gas; TGA: Thermal gravimetric analyser; TRL: Technical readiness level; TPP: Thermal power plants; WGS: Waste gas streams

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Data availability The datasets used and/or analyzed during the current study are available from the corresponding author on reasonable request.

Declarations

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Consent to participate Not applicable.

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References

- Abdalla M, Hastings A, Chadwick DR, Jones DL, Evans CD, Jones MB, Rees RM, Smith PETE (2018) Critical review of the impacts of grazing intensity on soil organic carbon storage and other soil quality indicators in extensively managed grasslands. *Agr Ecosyst Environ* 253:62–81
- Adu E, Zhang Y, Liu D (2019) Current situation of carbon dioxide capture, storage, and enhanced oil recovery in the oil and gas industry. *Canadian J Chem Eng* 97(5):1048–1076
- Al-Mamoori A, Krishnamurthy A, Rownaghi A & Rezaei F (2017) Carbon capture and utilization update. *Energ Technol* 5(6):834–849. In.
- Al-Shargabi M, Davoodi S, Wood DA, Rukavishnikov VS, Minaev KM (2022) Carbon dioxide applications for enhanced oil recovery assisted by nanoparticles: recent developments. *ACS Omega* 7(12):9984–9994
- Al Atrach J, Aitblal A, Amedlous A, Xiong Y, Desmurs M, Ruaux V, Guillet-Nicolas R, Valtchev V (2024) Nanosized zeolite P for enhanced CO₂ adsorption kinetics. *ACS Appl Mater Interfaces* 16(29):38006–38016
- Al Baroudi H, Awoyomi A, Patchigolla K, Jonnalagadda K, Anthony EJ (2021) A review of large-scale CO₂ shipping and marine emissions management for carbon capture, utilisation and storage. *Appl Energy* 287:116510
- Aldaco R, Butnar I, Margallo M, Laso J, Rumayor M, Dominguez-Ramos A, Irabien A, Dodds PE (2019) Bringing value to the chemical industry from capture, storage and use of CO₂: a dynamic LCA of formic acid production. *Sci Total Environ* 663:738–753
- Alexandra Petrovic B, Masoudi Soltani S (2019) Optimization of post combustion CO₂ capture from a combined-cycle gas turbine power plant via taguchi design of experiment. *Processes* 7(6):364
- Allen KA, Sikes EL, Hönisch B, Elmore AC, Guilderson TP, Rosenthal Y, Anderson RF (2015) Southwest Pacific deep water carbonate chemistry linked to high southern latitude climate and atmospheric CO₂ during the Last Glacial Termination. *Quatern Sci Rev* 122:180–191
- Alper E, Orhan OY (2017) CO₂ utilization: developments in conversion processes. *Petroleum* 3(1):109–126
- Aminu MD, Nabavi SA, Rochelle CA, Manovic V (2017) A review of developments in carbon dioxide storage. *Appl Energy* 208:1389–1419
- Amoakwah E, Arthur E, Frimpong KA, Parikh SJ, Islam R (2020) Soil organic carbon storage and quality are impacted by corn cob biochar application on a tropical sandy loam. *J Soils Sediments* 20:1960–1969
- Amouroux J, Siffert P, Massué JP, Cavadias S, Trujillo B, Hashimoto K, Rutberg P, Dresvin S, Wang X (2014) Carbon dioxide: a new material for energy storage. *Progress Natural Sci Mater Int* 24(4):295–304
- Anwar MN, Fayyaz A, Sohail NF, Khokhar MF, Baqar M, Khan WD, Rasool K, Rehan M, Nizami AS (2018) CO₂ capture and storage: a way forward for sustainable environment. *J Environ Manage* 226:131–144
- Aresta M, Dibenedetto A (2007) Utilisation of CO₂ as a chemical feedstock: opportunities and challenges. *Dalton Trans* 28:2975–2992
- Aresta M, Dibenedetto A, Angelini A (2014) Catalysis for the valorization of exhaust carbon: from CO₂ to chemicals, materials, and fuels. *Technological use of CO₂*. *Chem Rev* 114(3):1709–1742
- Azarabadi H, Lackner KS (2020) Postcombustion capture or direct air capture in decarbonizing US natural gas power? *Environ Sci Technol* 54(8):5102–5111
- Babaei M, Anbia M, Kazemipour M (2017) Improving CO₂ adsorption with new amine-functionalized Y-type zeolite. *J Adv Environ Health Res* 5(2):70–77
- Babakhani P, Phenrat T, Baalousha M, Soratana K, Peacock CL, Twinning BS, Hochella MF Jr (2022) Potential use of engineered nanoparticles in ocean fertilization for large-scale atmospheric carbon dioxide removal. *Nat Nanotechnol* 17(12):1342–1351
- Bacocchi R, Storti G, Mazzotti M (2006) Process design and energy requirements for the capture of carbon dioxide from air. *Chem Eng Process* 45(12):1047–1058
- Bahman N, Al-Khalifa M, Al Baharna S, Abdulmohsen Z, Khan E (2023) Review of carbon capture and storage technologies in selected industries: potentials and challenges. *Rev Environ Sci Bio/technol* 22(2):451–470
- Bai F, Jia C, Hu J, Alsousy A, Lu Y, Sepehrnoori K (2024) Storage capacity comparison of hydrogen and carbon dioxide in heterogeneous aquifers. *Gas Science and Engineering* 121:205182
- Bello OS, Adegoke KA, Olaniyan AA, Abdulazeez H (2015) Dye adsorption using biomass wastes and natural adsorbents: overview and future prospects. *Desalin Water Treat* 53(5):1292–1315
- Bhattacharyya D, Miller DC (2017) Post-combustion CO₂ capture technologies—a review of processes for solvent-based and sorbent-based CO₂ capture. *Curr Opin Chem Eng* 17:78–92
- Bhavsar A, Hingar D, Ostwal S, Thakkar I, Jadeja S, Shah M (2023) The current scope and stand of carbon capture storage and utilization~ a comprehensive review. *Case Studies Chem Environ Eng* 8:100368
- Blomen E, Hendriks C, Neele F (2009) Capture technologies: improvements and promising developments. *Energy Procedia* 1(1):1505–1512
- Bongole K, Sun Z, Yao J, Mehmood A, Yueying W, Mboje J, Xin Y (2019) Multifracture response to supercritical CO₂-EGS and water-EGS based on thermo-hydro-mechanical coupling method. *Int J Energy Res* 43(13):7173–7196
- Brack D, King R (2021) Managing land-based CDR: BECCS, forests and carbon sequestration. *Global Pol* 12:45–56
- Bui M, Adjiman CS, Bardow A, Anthony EJ, Boston A, Brown S, Fennell PS, Fuss S, Galindo A, Hackett LA, Hallett JP (2018) Carbon capture and storage (CCS): the way forward. *Energy Environ Sci* 11(5):1062–1176
- Burrows LC, Haeri F, Cvetic P, Sanguinito S, Shi F, Tapriyal D, Goodman A, Enick RM (2020) A literature review of CO₂, natural gas, and water-based fluids for enhanced oil recovery in unconventional reservoirs. *Energy Fuels* 34(5):5331–5380
- Cabral RP, Bui M, Mac Dowell N (2019) A synergistic approach for the simultaneous decarbonisation of power and industry

- via bioenergy with carbon capture and storage (BECCS). *Int J Greenhouse Gas Control* 87:221–237
- Caineng ZOU, Songtao WU, Zhi YANG, Songqi PAN, Guofeng WANG, Jiang X, Modi GUAN, Cong YU, Zhichao YU, Yue SHEN (2023) Progress, challenge and significance of building a carbon industry system in the context of carbon neutrality strategy. *Pet Explor Dev* 50(1):210–228
- Chang C-W, Kao Y-H, Shen P-H, Kang P-C, Wang C-Y (2020) Nano-confinement of metal oxide MgO and ZnO in zeolitic imidazolate framework ZIF-8 for CO₂ adsorption and regeneration. *J Hazard Mater* 400:122974
- Chapman M, Walker WS, Cook-Patton SC, Ellis PW, Farina M, Griscom BW, Baccini A (2020) Large climate mitigation potential from adding trees to agricultural lands. *Glob Change Biol* 26(8):4357–4365
- Chea C, Ket P, Taing L, Kong S, Um D, Taing C, Or C, Aun S, Hang L (2022) Life-cycle impact assessment of air emissions from a cement production plant in Cambodia. *Open Journal of Air Pollution* 11(4):85–99
- Cheah WY, Ling TC, Juan JC, Lee DJ, Chang JS, Show PL (2016) Biorefineries of carbon dioxide: from carbon capture and storage (CCS) to bioenergies production. *Biores Technol* 215:346–356
- Chen S, Liu J, Zhang Q, Teng F, McLellan BC (2022) A critical review on deployment planning and risk analysis of carbon capture, utilization, and storage (CCUS) toward carbon neutrality. *Renew Sustain Energy Rev* 167:112537
- Chen S, Yu R, Soomro A, Xiang W (2019) Thermodynamic assessment and optimization of a pressurized fluidized bed oxy-fuel combustion power plant with CO₂ capture. *Energy* 175:445–455
- Choi YH, Jang YJ, Park H, Kim WY, Lee YH, Choi SH, Lee JS (2017) Carbon dioxide Fischer-Tropsch synthesis: a new path to carbon-neutral fuels. *Appl Catal B* 202:605–610
- Connelly DP, Bull JM, Flohr A, Schaap A, Koopmans D, Blackford JC, White PR, James RH, Pearce C, Lichtschlag A, Achterberg EP (2022) Assuring the integrity of offshore carbon dioxide storage. *Renew Sustain Energy Rev* 166:112670
- Czarnota R, Janiga D, Stopa J, Wojnarowski P (2017) Determination of minimum miscibility pressure for CO₂ and oil system using acoustically monitored separator. *J CO₂ Utilization* 17:32–36
- Damen K, van Troost M, Faaij A, Turkenburg W (2006) A comparison of electricity and hydrogen production systems with CO₂ capture and storage Part A: review and selection of promising conversion and capture technologies. *Progress Energy Combustion Sci* 32(2):215–246
- Davoodi S, Al-Shargabi M, Wood DA, Mehrad M, Rukavishnikov VS (2024) Carbon dioxide sequestration through enhanced oil recovery: a review of storage mechanisms and technological applications. *Fuel* 366:131313
- Davoodi S, Al-Shargabi M, Wood DA, Rukavishnikov VS, & Minaev KM (2023) Review of technological progress in carbon dioxide capture, storage, and utilization. *Gas Science and Engineering* 205070.
- Decarre S, Berthiaud J, Butin N, Guillaume-Combecave JL (2010) CO₂ maritime transportation. *Int J Greenhouse Gas Control* 4(5):857–864
- Dinca C, Slavu N, Badea A (2018a) Benchmarking of the pre/post-combustion chemical absorption for the CO₂ capture. *J Energy Inst* 91(3):445–456
- Dinca C, Slavu N, Cormos CC, Badea A (2018b) CO₂ capture from syngas generated by a biomass gasification power plant with chemical absorption process. *Energy* 149:925–936
- Dissanayake PD, Choi SW, Igalavithana AD, Yang X, Tsang DC, Wang CH, Kua HW, Lee KB, Ok YS (2020) Sustainable gasification biochar as a high efficiency adsorbent for CO₂ capture: a facile method to designer biochar fabrication. *Renew Sustain Energy Rev* 124:109785
- Douskova I, Doucha J, Livansky K, Machat J, Novak P, Umysova D, Zachleder V, Vitova M (2009) Simultaneous flue gas bioremediation and reduction of microalgal biomass production costs. *Appl Microbiol Biotechnol* 82:179–185
- Dziejarski B, Krzyżyńska R, Andersson KJF (2023) Current status of carbon capture, utilization, and storage technologies in the global economy: a survey of technical assessment. *Fuel* 342:27776
- El Nasr AS, Nelson T, Kataria A, Abu-Zahra MR (2015) Benchmarking of a novel solid sorbent CO₂ capture process for NGCC power generation. *Int J Greenhouse Gas Control* 42:583–592
- Elhenawy SEM, Khraisheh M, AlMomani F, Walker G (2020) Metal-organic frameworks as a platform for CO₂ capture and chemical processes: adsorption, membrane separation, catalytic-conversion, and electrochemical reduction of CO₂. *Catalysts* 10(11):1293
- Emrich W (2013) *Handbook of charcoal making: the traditional and industrial methods* (Vol. 7): Springer Science & Business Media.
- Engel F, Kather A (2018) Improvements on the liquefaction of a pipeline CO₂ stream for ship transport. *Int J Greenhouse Gas Control* 72:214–221
- Esiri AE, Jambol DD, Ozowe C (2024) Best practices and innovations in carbon capture and storage (CCS) for effective CO₂ storage. *Int Journal Appl Res Social Sci* 6(6):1227–1243
- Fan W, Xiao C, Li P, Zhang Z, Lin T, Pan Y, Di Y, Chen Y (2020) Intelligent control system of an ecological engineering project for carbon sequestration in coastal mariculture environments in China. *Sustainability* 12(13):5227
- Fashi F, Ghaemi A, Behrooz AH (2021) Piperazine impregnation on Zeolite 13X as a novel adsorbent for CO₂ capture: experimental and modeling. *Chem Eng Commun* 208(8):1104–1120
- Fasihi M, Efimova O, Breyer C (2019) Techno-economic assessment of CO₂ direct air capture plants. *J Clean Prod* 224:957–980
- Fei Z, Alshuraiaan B, Abed AM, Wae-hayee M (2023) CO₂ capture and liquefaction in a novel zero carbon emission co-production scheme for ships by integration of supercritical CO₂-based oxy-fuel cycle and ion transport membrane; Energy/exergy/exergo-economic/exergoenvironmental (4E) and multi-objective optimization. *Sep Purif Technol* 309:123037
- Feron PH, Cousins A, Gao S, Liu L, Wang J, Wang S, Niu H, Yu H, Li K, Cottrell A (2017) Experimental performance assessment of a mono-ethanolamine-based post-combustion CO₂-capture at a coal-fired power station in China. *Greenhouse Gases: Sci Technol* 7(3):486–499
- Figuerola JD, Fout T, Plasynski S, McIlvried H, Srivastava RD (2008) Advances in CO₂ capture technology—the US Department of Energy’s Carbon Sequestration Program. *Int J Greenhouse Gas Control* 2(1):9–20
- Friedmann SJ (2008) Carbon dioxide sequestration in geological media: state of the science. *Environ Geosci* 15:63–77. <https://doi.org/10.1306/eg.11230707015>
- Fuss S, Lamb WF, Callaghan MW, Hilaire J, Creutzig F, Amann T, Beringer T, de Oliveira Garcia W, Hartmann J, Khanna T, Luderer G (2018) Negative emissions—part 2: costs, potentials and side effects. *Environ Res Lett* 13(6):063002
- Gao G, Jiang W, Li X, Zhao Z, Jiang C, Luo C, Wu F, Zhang L (2023) Novel assessment of highly efficient polyamines for post-combustion CO₂ capture: absorption heat, reaction rate, CO₂ cyclic capacity, and phase change behavior. *Sep Purif Technol* 306:122615
- Ghezlbash A, Khaligh V, Fahimifard SH, Liu JJ (2023) A comparative perspective of the effects of CO₂ and non-CO₂ greenhouse gas emissions on global solar, wind, and geothermal energy investment. *Energies* 16(7):3025
- Ghosh A, Kiran B (2017) Carbon concentration in algae: reducing CO₂ from exhaust gas. *Trends Biotechnol* 35(9):806–808

- Godin J, Liu W, Ren S, Xu CC (2021) Advances in recovery and utilization of carbon dioxide: a brief review. *J Environ Chem Eng* 9(4):105644
- Gowd SC, Ganeshan P, Vigneswaran VS, Hossain MS, Kumar D, Rajendran K, Ngo HH, Pugazhendhi A (2023) Economic perspectives and policy insights on carbon capture, storage, and utilization for sustainable development. *Sci Total Environ* 883:163656
- Guo S, Li Y, Wang Y, Wang L, Sun Y, & Liu L (2022) Recent advances in biochar-based adsorbents for CO₂ capture. *Carbon Capture Science & Technology*, 100059.
- Gupta S, Kua HW, Low CY (2018) Use of biochar as carbon sequestering additive in cement mortar. *Cement Concr Compos* 87:110–129
- Guruviah KD, Sivasankaran C, & Bharathiraja B (2019) Thermochemical conversion: bio-oil and syngas production. In *Prospects of Renewable Bioprocessing in Future Energy Systems* (pp. 251–267): Springer.
- Hamza A, Hussein IA, Al-Marri MJ, Mahmoud M, Shawabkeh R, Aparicio S (2021) CO₂ enhanced gas recovery and sequestration in depleted gas reservoirs: a review. *J Petrol Sci Eng* 196:107685
- Hasan MR, Khodadad Mostakim MSI, Amir NB, & Taufique Ahmmed MRA (2020) *Municipal Solid Waste Management: Scopes, Challenges of Sustainability and Treatments in Rajshahi City, Bangladesh*. Paper presented at the Proc. Int. Conf. Mech., Ind., & Energy Eng., Bangladesh.
- Hepburn C, Adlen E, Beddington J, Carter EA, Fuss SN, Dowell M, Minx JC, Smith P, Williams CK (2019) The technological and economic prospects for CO₂ utilization and removal. *Nature* 575:87–97
- Hong, WY J. C. C. S. & Technology (2022) A techno-economic review on carbon capture, utilisation and storage systems for achieving a net-zero CO₂ emissions future 3:100044.
- Hospital-Benito D, Lemus J, Moya C, Santiago R, Paramio C, Palomar J (2022) Aspen plus supported design of pre-combustion CO₂ capture processes based on ionic liquids. *Sep Purif Technol* 290:120841
- Huang P-H, Cheng H-H & Lin S-H (2015) Adsorption of carbon dioxide onto activated carbon prepared from coconut shells. *Journal of Chemistry*, 2015.
- Huang X, Ai N, Li L, Jiang Q, Wang Q, Ren J, Wang J (2022) Simulation of CO₂ capture process in flue gas from oxy-fuel combustion plant and effects of properties of absorbent. *Separations* 9(4):95
- Hurd CL, Law CS, Bach LT, Britton D, Hovenden M, Paine ER, Raven JA, Tamsitt V, Boyd PW (2022) Forensic carbon accounting: assessing the role of seaweeds for carbon sequestration. *J Phycol* 58(3):347–363
- Jessop PG (2011) Searching for green solvents. *Green Chemistry* 13(6). <https://doi.org/10.1039/c0gc00797h>
- Joarder M, Rashid F & Joarder T (2019) *Solution to air pollution for removing CO₂ and SO₂ from flue gases: a prospective approach*. Paper presented at the International Conference on Mechanical Engineering and Renewable Energy.
- Johnsen K, Helle K, Rønneid S, Holt H (2011) DNV recommended practice: Design and operation of CO₂ pipelines. *Energy Procedia* 4:3032–3039
- Kalair A, Abas N, Saleem MS, Kalair AR, Khan NJES (2021) Role of Energy Storage Systems in Energy Transition from Fossil Fuels to Renewables 3(1):e135
- Kato M, Maezawa Y, Takeda S, Hagiwara Y, Kogo R, Semba K & Hamamura MJ. K. E. M. (2006). Pre-combustion CO₂ capture using ceramic absorbent and methane steam reforming 317:81–84.
- Kazemifar F (2022) A review of technologies for carbon capture, sequestration, and utilization: cost, capacity, and technology readiness. *Greenhouse Gases: Sci Technol* 12(1):200–230
- Keith DW, Holmes G, Angelo DS, Heidel K (2018) A process for capturing CO₂ from the atmosphere. *Joule* 2(8):1573–1594
- Kenarsari SD, Yang D, Jiang G, Zhang S, Wang J, Russell AG, Wei Q, Fan M (2013) Review of recent advances in carbon dioxide separation and capture. *RSC Adv* 3(45):22739–22773
- Khan IU, Othman MHD, Jilani A, Ismail AF, Hashim H, Jaafar J, Rahman MA, Rehman GU (2018a) Economical, environmental friendly synthesis, characterization for the production of zeolitic imidazolate framework-8 (ZIF-8) nanoparticles with enhanced CO₂ adsorption. *Arab J Chem* 11(7):1072–1083
- Khan MA, Ngo HH, Guo W, Liu Y, Zhang X, Guo J, Chang SW, Nguyen DD, Wang J (2018b) Biohydrogen production from anaerobic digestion and its potential as renewable energy. *Renewable Energy* 129:754–768
- Kim S, Ko Y, Lee GJ, Lee JW, Xu R, Ahn H, Kang YT (2023) Sustainable energy harvesting from post-combustion CO₂ capture using amine-functionalized solvents. *Energy* 267:126532
- Koytsoumpa EI, Bergins C, Kakaras E (2018) The CO₂ economy: review of CO₂ capture and reuse technologies. *The Journal of Supercritical Fluids* 132:3–16
- Kumar R, Mangalapuri R, Ahmadi MH, Vo D-VN, Solanki R, Kumar P (2020a) The role of nanotechnology on post-combustion CO₂ absorption in process industries. *International Journal of Low-Carbon Technologies* 15(3):361–367
- Kumar S, Foroozesh J, Edlmann K, Rezk MG, Lim CY (2020b) A comprehensive review of value-added CO₂ sequestration in subsurface saline aquifers. *Journal of Natural Gas Science and Engineering* 81:103437
- Lam MK, Lee KT, Mohamed AR (2012) Current status and challenges on microalgae-based carbon capture. *Int J Greenhouse Gas Control* 10:456–469
- Lamm B (2019) Algae might be a secret weapon to combatting climate change. *Quartz* 30:722–727
- Lau HC, Ramakrishna S, Zhang K, Radhamani AV (2021) The role of carbon capture and storage in the energy transition. *Energy Fuels* 35(9):7364–7386
- Lee CT, Mohammad Rozali NE, Van Fan Y, Klemeš JJ, Towprayoon S (2018) Low-carbon emission development in Asia: energy sector, waste management and environmental management system. *Clean Technol Environ Policy* 20:443–449
- Leung DY, Caramanna G, Maroto-Valer MM (2014) An overview of current status of carbon dioxide capture and storage technologies. *Renew Sustain Energy Rev* 39:426–443
- Li-ping H, Ping-ping S, Xin-wei L, Qi-Chao G, Cheng-sheng W, Fang-fang L (2015) Study on CO₂ EOR and its geological sequestration potential in oil field around Yulin city. *J Petrol Sci Eng* 134:199–204
- Liu E, Lu X, Wang D (2023) A systematic review of carbon capture, utilization and storage: status, progress and challenges. *Energies* 16(6):2865
- Liu Q, Wu L, Jackstell R & Beller M (2015) Using carbon dioxide as a building block in organic synthesis. *Nat Commun* 6:5933. Retrieved from <https://www.ncbi.nlm.nih.gov/pubmed/25600683>. <https://doi.org/10.1038/ncomms6933>
- Liu W, Ramirez A (2017) State of the art review of the environmental assessment and risks of underground geo-energy resources exploitation. *Renew Sustain Energy Rev* 76:628–644
- Liu W, Teng L, Rohani S, Qin Z, Zhao B, Xu CC, Ren S, Liu Q, Liang B (2021) CO₂ mineral carbonation using industrial solid wastes: a review of recent developments. *Chem Eng J* 416:129093
- Liu Y, Rui Z (2022) A storage-driven CO₂ EOR for a net-zero emission target. *Engineering* 18:79–87

- Loria P, Bright MB (2021) Lessons captured from 50 years of CCS projects. *Electr J* 34(7):106998
- Lu H, Ma X, Huang K, Fu L, Azimi M (2020) Carbon dioxide transport via pipelines: A systematic review. *J Clean Prod* 266:121994
- Luo J, Xie Y, Hou MZ, Xiong Y, Wu X, Lüddecke CT, Huang LJER (2023) Advances in Subsea Carbon Dioxide Utilization and Storage 2(1):100016
- Malekli M, Aslani A (2022) A novel post-combustion CO₂ capture design integrated with an Organic Rankine Cycle (ORC). *Process Saf Environ Prot* 168:942–952
- Manceau JC, Hatzignatiou DG, De Lary L, Jensen NB, Réveillère A (2014) Mitigation and remediation technologies and practices in case of undesired migration of CO₂ from a geological storage unit—current status. *Int J Greenhouse Gas Control* 22:272–290
- Marriott R, Jessop P & Barnes M (2015) CO₂-based solvents. In *Carbon Dioxide Utilisation* (pp. 73–96): Elsevier.
- Martyn Poliakoff JMF, Farren TR, Anastas' PT (2002) Green chemistry: science and politics of change. *Science* 297:807–810. 1010
- Mathimani T, Baldinelli A, Rajendran K, Prabakar D, Matheswaran M, van Leeuwen RP, Pugazhendhi A (2019) Review on cultivation and thermochemical conversion of microalgae to fuels and chemicals: process evaluation and knowledge gaps. *J Clean Prod* 208:1053–1064
- McLaughlin H, Littlefield AA, Menefee M, Kinzer A, Hull T, Sovacool BK, Bazilian MD, Kim J, Griffiths S (2023) Carbon capture utilization and storage in review: sociotechnical implications for a carbon reliant world. *Renew Sustain Energy Rev* 177:113215
- McQueen N, Gomes KV, McCormick C, Blumanthal K, Pisciotta M, Wilcox J (2021) A review of direct air capture (DAC): scaling up commercial technologies and innovating for the future. *Progress in Energy* 3(3):032001
- Memmah MM, Lescourret F, Yao X, Lavigne C (2015) Metaheuristics for agricultural land use optimization. *A Review Agronomy for Sustainable Development* 35:975–998
- Mikulčić H, Skov IR, Dominković DF, Alwi SRW, Manan ZA, Tan R, Duić N, Mohamad SNH, Wang X (2019) Flexible carbon capture and utilization technologies in future energy systems and the utilization pathways of captured CO₂. *Renew Sustain Energy Rev* 114:109338
- Mogensen K, Masalmeh S (2020) A review of EOR techniques for carbonate reservoirs in challenging geological settings. *J Petrol Sci Eng* 195:107889
- Mondal MK, Balsora HK, Varshney P (2012) Progress and trends in CO₂ capture/separation technologies: a review. *Energy* 46(1):431–441
- Morecroft MD, Duffield S, Harley M, Pearce-Higgins JW, Stevens N, Watts O & Whitaker JJS (2019) Measuring the success of climate change adaptation and mitigation in terrestrial ecosystems. 366(6471):eaaw9256.
- Munda S, Bhaduri D, Mohanty S, Chatterjee D, Tripathi R, Shahid M, Kumar U, Bhattacharyya P, Kumar A, Adak T, Jangde HK (2018) Dynamics of soil organic carbon mineralization and C fractions in paddy soil on application of rice husk biochar. *Biomass Bioenerg* 115:1–9
- Nandakishora Y, Sahoo R & Murugan S (2021) *Review on waste heat recovery from flue gas and its application in CO₂ capture*. Paper presented at the IOP Conference Series: Materials Science and Engineering.
- Nave LE, Walters BF, Hofmeister KL, Perry CH, Mishra U, Domke GM, Swanston CW (2019) The role of reforestation in carbon sequestration. *New Forest* 50(1):115–137
- Nolan CJ, Field CB, Mach KJ (2021) Constraints and enablers for increasing carbon storage in the terrestrial biosphere. *Nature Reviews Earth & Environment* 2(6):436–446
- Nothout P, Wiersma F, Hurtado O, Macdonald D, Kemper J, van Alphen K (2014) CO₂ Pipeline infrastructure—lessons learnt. *Energy Procedia* 63:2481–2492
- Northrup DL, Basso B, Wang MQ, Morgan CL, Benfey PN (2021) Novel technologies for emission reduction complement conservation agriculture to achieve negative emissions from row-crop production. *Proc Natl Acad Sci* 118(28):e2022666118
- Nunes LJ, Meireles CI, Pinto Gomes CJ, Almeida Ribeiro NM (2019) Forest management and climate change mitigation: a review on carbon cycle flow models for the sustainability of resources. *Sustainability* 11(19):5276
- Olajire AA (2010) CO₂ capture and separation technologies for end-of-pipe applications—a review. *Energy* 35(6):2610–2628
- Omari A, Wang C, Li Y, Xu X (2022) The progress of enhanced gas recovery (EGR) in shale gas reservoirs: a review of theory, experiments, and simulations. *J Petrol Sci Eng* 213:110461
- Omodolor IS, Otor HO, Andonegui JA, Allen BJ, Alba-Rubio AC (2020) Dual-function materials for CO₂ capture and conversion: a review. *Ind Eng Chem Res* 59(40):17612–17631
- Osman AI, Deka TJ, Baruah DC & Rooney DW (2020) Critical challenges in biohydrogen production processes from the organic feedstocks. *Biomass Conversion and Biorefinery* 1–19.
- Osman AI, Hefny M, Abdel Maksoud M, Elgarahy AM, Rooney DW (2021) Recent advances in carbon capture storage and utilisation technologies: a review. *Environ Chem Lett* 19(2):797–849
- Paluszny A, Graham CC, Daniels KA, Tsaparlis V, Xenias D, Salimzadeh S, Whitmarsh L, Harrington JF, Zimmerman RW (2020) Caprock integrity and public perception studies of carbon storage in depleted hydrocarbon reservoirs. *Int J Greenhouse Gas Control* 98:103057
- Pan B, Yin X, Iglauer S (2021) Rock-fluid interfacial tension at sub-surface conditions: implications for H₂, CO₂ and natural gas geo-storage. *Int J Hydrogen Energy* 46(50):25578–25585
- Pan B, Yin X, Zhu W, Yang Y, Ju Y, Yuan Y, Iglauer S (2022) Theoretical study of brine secondary imbibition in sandstone reservoirs: implications for H₂, CH₄, and CO₂ geo-storage. *Int J Hydrogen Energy* 47(41):18058–18066
- Park J, Cho SY, Jung M, Lee K, Nah Y-C, Attia NF, & Oh H (2021) Efficient synthetic approach for nanoporous adsorbents capable of pre-and post-combustion CO₂ capture and selective gas separation. *Journal of CO₂ Utilization*, 45:101404.
- Perpiñán J, Bailera M, Romeo LM, Peña B, Evely V (2021) CO₂ recycling in the iron and steel industry via power-to-gas and oxy-fuel combustion. *Energies* 14(21):7090
- Portillo E, Alonso-Fariñas B, Vega F, Cano M, Navarrete B (2019) Alternatives for oxygen-selective membrane systems and their integration into the oxy-fuel combustion process: A review. *Sep Purif Technol* 229:115708
- Pregitzer CC, Hanna C, Charlop-Powers S, Bradford MA (2022) Estimating carbon storage in urban forests of New York City. *Urban Ecosystems* 25(2):617–631
- Psarras PC, Comello S, Bains P, Charoensawadpong P, Reichelstein S, Wilcox J (2017) Carbon capture and utilization in the industrial sector. *Environ Sci Technol* 51(19):11440–11449
- Raganati F, Ammendola P (2024) CO₂ post-combustion capture: a critical review of current technologies and future directions. *Energy Fuels* 38(15):13858–13905
- Rahman FA, Aziz MMA, Saidur R, Bakar WAWA, Hainin M, Putrajaya R, Hassan NA (2017) Pollution to solution: capture and sequestration of carbon dioxide (CO₂) and its utilization as a renewable energy source for a sustainable future. *Renew Sustain Energy Rev* 71:112–126
- Raman R (2016) Advances in microalgae engineering and synthetic biology applications for biofuel production. *Curr Opin Chem Biol* 34:60–66

- Rau GH, Baird JR (2018) Negative- CO_2 -emissions ocean thermal energy conversion. *Renew Sustain Energy Rev* 95:265–272
- Raza A, Glatz G, Gholami R, Mahmoud M, Alafnan S (2022) Carbon mineralization and geological storage of CO_2 in basalt: mechanisms and technical challenges. *Earth Sci Rev* 229:104036
- Razzak SA, Hossain MM, Lucky RA, Bassi AS, De Lasa H (2013) Integrated CO_2 capture, wastewater treatment and biofuel production by microalgae culturing—a review. *Renew Sustain Energy Rev* 27:622–653
- Reverchon E, De Marco I (2006) Supercritical fluid extraction and fractionation of natural matter. *The Journal of Supercritical Fluids* 38(2):146–166. <https://doi.org/10.1016/j.supflu.2006.03.020>
- Roussanal S, Hognes ES, Jakobsen JP (2013) Multi-criteria analysis of two CO_2 transport technologies. *Energy Procedia* 37:2981–2988
- Rubin ES, Davison JE, Herzog HJ (2015) The cost of CO_2 capture and storage. *Int J Greenhouse Gas Control* 40:378–400
- Sanchez DL, Johnson N, McCoy ST, Turner PA, Mach KJ (2018) Near-term deployment of carbon capture and sequestration from biorefineries in the United States. *Proc Natl Acad Sci* 115(19):4875–4880
- Sanz-Pérez ES, Murdock CR, Didas SA, Jones CW (2016) Direct capture of CO_2 from ambient air. *Chem Rev* 116(19):11840–11876
- Sayma AI, Morsi BI, Lee M, Dryer FL (2014) Numerical investigation of catalytic methanation of carbon dioxide for methane production and storage. *Energy Fuels* 28(3):2173–2185
- Senior CL, Morris W, Lewandowski TA (2013) Emissions and risks associated with oxyfuel combustion: state of the science and critical data gaps. *J Air Waste Manag Assoc* 63(7):832–843
- Shang WL, Lv Z (2023) Low carbon technology for carbon neutrality in sustainable cities: a survey. *Sustain Cities Soc* 92:104489
- Sharma V, Dash S, Gupta P (2024) Comprehensive review of fly ash: environmental impact and applications. *Environ Qual Manage* 34(2):e22338
- Sher F, Iqbal SZ, Albazzaz S, Ali U, Mortari DA, Rashid T (2020) Development of biomass derived highly porous fast adsorbents for post-combustion CO_2 capture. *Fuel* 282:118506
- Sinha A, Realff MJ (2019) A parametric study of the techno-economics of direct CO_2 air capture systems using solid adsorbents. *AIChE J* 65(7):e16607
- Smith K (2019) Fuel synthesis from captured CO_2 . *Nat Rev Chem* 3(9):498–500
- Snæbjörnsdóttir SÓ, Gislason SR, Galeczka IM, Oelkers EH (2018) Reaction path modelling of in-situ mineralisation of CO_2 at the CarbFix site at Hellisheidi, SW-Iceland. *Geochim Cosmochim Acta* 220:348–366
- Snæbjörnsdóttir SÓ, Sigfússon B, Marieni C, Goldberg D, Gislason SR, Oelkers EH (2020) Carbon dioxide storage through mineral carbonation. *Nature Reviews Earth & Environment* 1(2):90–102
- Sohaib Q, Muhammad A, Younas M, Rezakazemi M (2020) Modeling pre-combustion CO_2 capture with tubular membrane contactor using ionic liquids at elevated temperatures. *Sep Purif Technol* 241:116677
- Sohaib Q, Muhammad A, Younas M, Rezakazemi M, Druon-Bocquet S, Sanchez-Marciano J (2021) Rigorous non-isothermal modeling approach for mass and energy transport during CO_2 absorption into aqueous solution of amino acid ionic liquids in hollow fiber membrane contactors. *Sep Purif Technol* 254:117644
- Steenveeldt R, Berger B, Torp T (2006) CO_2 capture and storage: closing the knowing–doing gap. *Chem Eng Res des* 84(9):739–763
- Subraveti SG, Pai KN, Rajagopalan AK, Wilkins NS, Rajendran A, Jayaraman A, Alptekin G (2019) Cycle design and optimization of pressure swing adsorption cycles for pre-combustion CO_2 capture. *Appl Energy* 254:113624
- Sun C, Liu S, Li J, Zeng H, Luo JL (2019) Insights into the interfacial process in electroless Ni–P coating on supercritical CO_2 transport pipeline as relevant to carbon capture and storage. *ACS Appl Mater Interfaces* 11(17):16243–16251
- Teng L, Liu X, Li X, Li Y, Lu C (2021) An approach of quantitative risk assessment for release of supercritical CO_2 pipelines. *Journal of Natural Gas Science and Engineering* 94:104131
- Thonemann N, Pizzol M (2019) Consequential life cycle assessment of carbon capture and utilization technologies within the chemical industry. *Energy Environ Sci* 12(7):2253–2263
- Tong D, Zhang Q, Zheng Y, Caldeira K, Shearer C, Hong C, Qin Y and Davis SJ 2019 Committed emissions from existing energy infrastructure jeopardize 1.5 C climate target. *Nature* 572(7769):373–377.
- Tushar MSH, Islam MS, Ahmmed T & Joarder MSA (2024) Simulation of biocrude production from P. Tricornutum, S. Platensis, and C. Vulgaris Using Aspenplus®. *Heliyon* 10(17).
- Usman M, Iqbal N, Noor T, Zaman N, Asghar A, Abdelnaby MM, Galadima A, Helal A (2022) Advanced strategies in metal-organic frameworks for CO_2 capture and separation. *Chem Rec* 22(7):e202100230
- Valluri S, Claremboux V, Kawatra S (2022) Opportunities and challenges in CO_2 utilization. *J Environ Sci* 113:322–344
- Vega F, Camino S, Camino J, Garrido J & Navarrete BJ. A. E. (2019). Partial oxy-combustion technology for energy efficient CO_2 capture process 253:113519.
- Vega F, Cano M, Gallego LM, Camino S, Camino JA, Navarrete B (2017) Evaluation of MEA 5 M performance at different CO_2 concentrations of flue gas tested at a CO_2 capture lab-scale plant. *Energy Procedia* 114:6222–6228
- Wang L, Deng J, Yang X, Hou R, Hou D (2023) Role of biochar toward carbon neutrality. *Carbon Research* 2(1):2
- Wang X, Song C (2020) Carbon capture from flue gas and the atmosphere: a perspective. *Frontiers in Energy Research* 8:560849
- Wang Y, Zhao L, Otto A, Robinius M, Stolten D (2017) A review of post-combustion CO_2 capture technologies from coal-fired power plants. *Energy Procedia* 114:650–665
- Wei X, Manovic V, Hanak DP (2020) Techno-economic assessment of coal-or biomass-fired oxy-combustion power plants with supercritical carbon dioxide cycle. *Energy Convers Manage* 221:113143
- Williams A, Jones J, Ma L, Pourkashanian M (2012) Pollutants from the combustion of solid biomass fuels. *Prog Energy Combust Sci* 38(2):113–137
- Woodall CM, McQueen N, Pilorgé H, Wilcox J (2019) Utilization of mineral carbonation products: current state and potential. *Greenhouse Gases: Science and Technology* 9(6):1096–1113
- Wu H-B, Xu M-X, Li Y-B, Wu J-H, Shen J-C, Liao H (2020) Experimental research on the process of compression and purification of CO_2 in oxy-fuel combustion. *Appl Energy* 259:114123
- Wu H & Duan QJS (2019). Gas void fraction measurement of gas-liquid two-phase CO_2 flow using laser attenuation technique. 19(14):3178.
- Xu X, Saeedi A, Liu K (2017) An experimental study of combined foam/surfactant polymer (SP) flooding for carbene dioxide-enhanced oil recovery (CO_2 -EOR). *J Petrol Sci Eng* 149:603–611
- Yave W, Car A, Wind J, Peinemann K-V (2010) Nanometric thin film membranes manufactured on square meter scale: ultra-thin films for CO_2 capture. *Nanotechnology* 21(39):395301
- Yuan L, Xu YJ (2015) Photocatalytic conversion of CO_2 into value-added and renewable fuels. *Appl Surf Sci* 342:154–167
- Yuan Q, Zhang Y, Wang T, Wang J, Romero CE (2022) Mineralization characteristics of coal fly ash in the transition from non-supercritical CO_2 to supercritical CO_2 . *Fuel* 318:123636
- Zhang N, Chai YE, Santos RM, Šiller L (2020) Advances in process development of aqueous CO_2 mineralisation towards scalability. *J Environ Chem Eng* 8(6):104453
- Zhang X, Qian C, Ma Z, Li F (2022) Study on preparation of supplementary cementitious material using microbial CO_2 fixation of steel slag powder. *Constr Build Mater* 326:126864

- Zhang Z, Huisingh D (2017) Carbon dioxide storage schemes: technology, assessment and deployment. *J Clean Prod* 142:1055–1064
- Zhao K, Jia C, Li Z, Du X, Wang Y, Li J, Yao Z, Yao J (2023) Recent advances and future perspectives in carbon capture, transportation, utilization, and storage (CCTUS) technologies: a comprehensive review. *Fuel* 351:128913
- Zhao L, Riensche E, Menzer R, Blum L, Stolten D (2008) A parametric study of CO₂/N₂ gas separation membrane processes for post-combustion capture. *J Membr Sci* 325(1):284–294
- Zhao N, Xu T, Wang K, Tian H, Wang F (2018) Experimental study of physical-chemical properties modification of coal after CO₂ sequestration in deep unmineable coal seams. *Greenhouse Gases: Science and Technology* 8(3):510–528
- Zhu Q (2019) Developments on CO₂-utilization technologies. *Clean Energy* 3(2):85–100
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